



WELCOME





Building Social Media Content Automation Workflow

Recap

Large Language Model (LLM)

It is an advanced type of AI model **trained on vast amounts of text** data to understand, generate, and process human language

Super Intelligent Model

Large

Language

Model

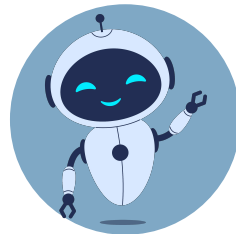
Generative AI Prompt

The natural language text, which **describes the task that an AI should perform**, is called a prompt



Describe the country “India” in two lines

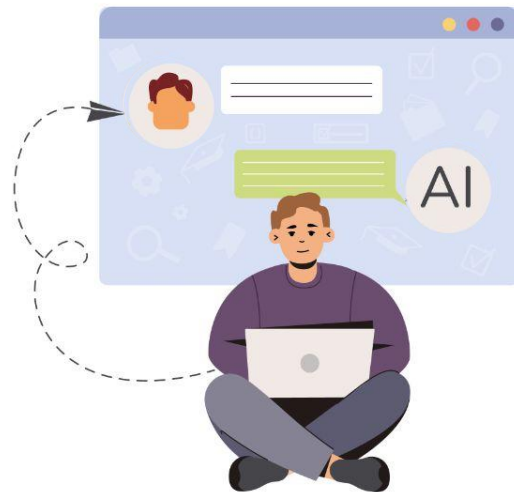
India is a vast South Asian country known for its incredibly diverse culture, languages, and religions. It is also the world's most populous democracy, characterized by a rich history and rapid economic growth



Recap

Prompt Engineering

Prompt engineering is the **art of creating prompts** that help an LLM generate more accurate and higher quality answers



Structuring Prompt

We can structure a prompt using three different roles



System

Sets the overall behavior, guidelines, and context for the AI



User

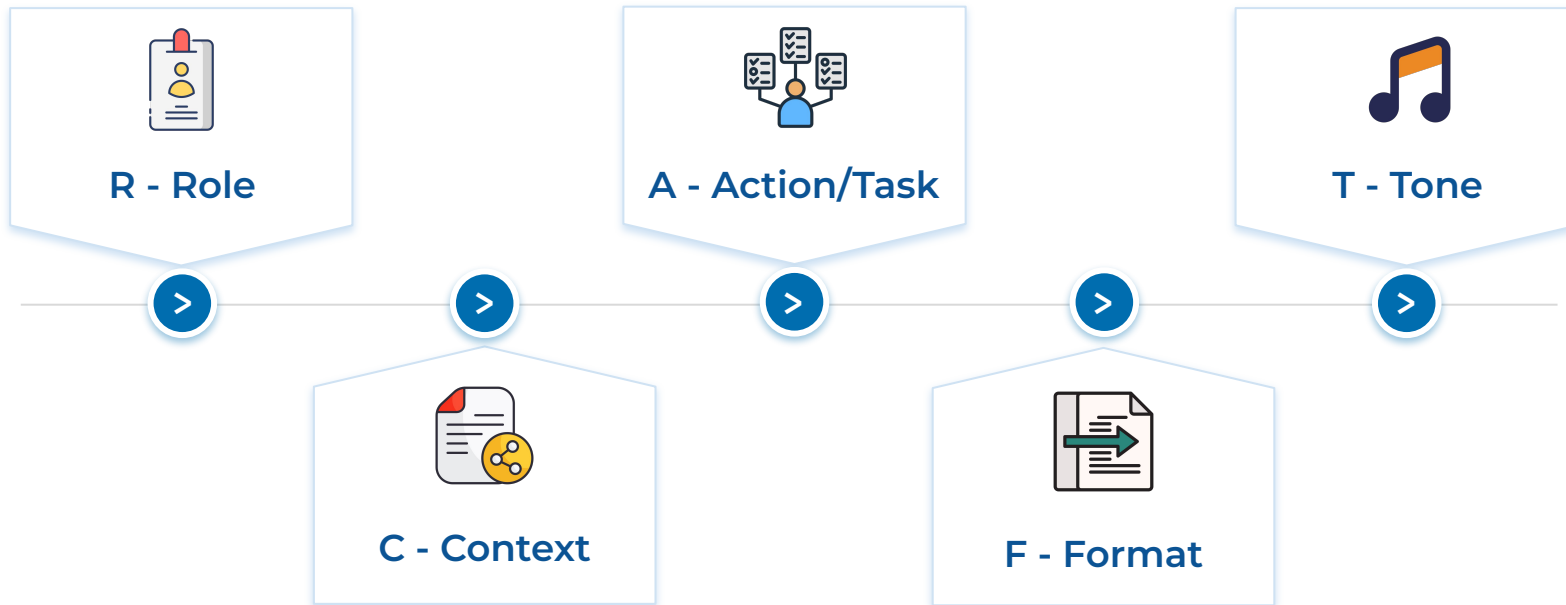
Provides the questions or tasks to the AI



Assistant

The AI's response based on the user's instructions

Basic Prompt Structure



Prompting Techniques

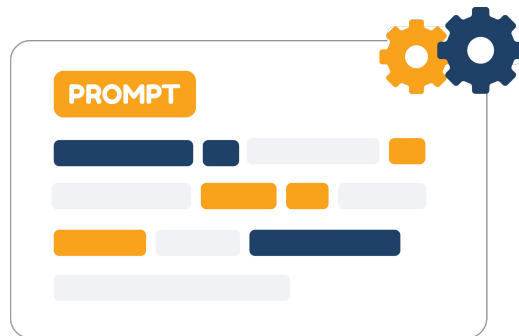
Some well-known prompting techniques are

✓ **Zero-shot**

✓ **One-shot**

✓ **Few-shot**

Many more ...



Agenda

AI Workflows & Automation

Introduction

Use Cases



AI Workflows

AI-Powered Social Media
Content Automation

Quick Question

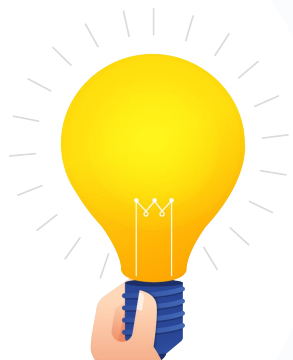


Have you ever spent too much time posting
on social media about your learning
progress/edu/career updates?

(writing explanations, adding tech hashtags,
formatting, posting, etc)



Quick Question



What if you had a system that could handle all of this stuff automatically?



AI-Powered Social Media Content Automation



AI Workflows



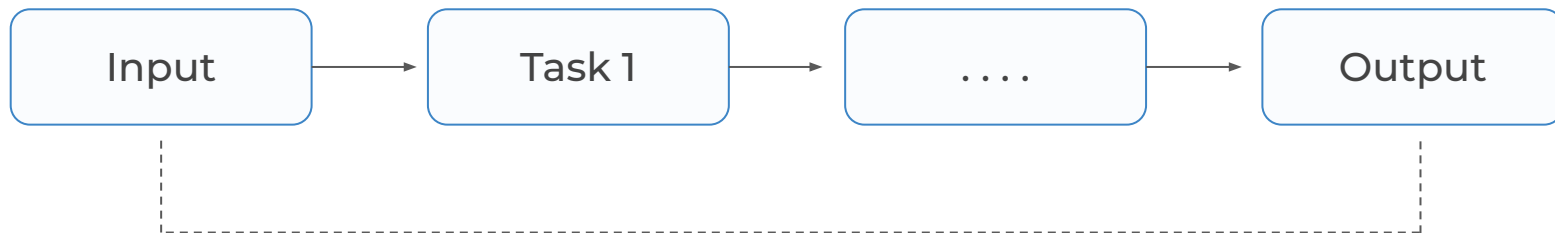
Introduction

AI Workflows

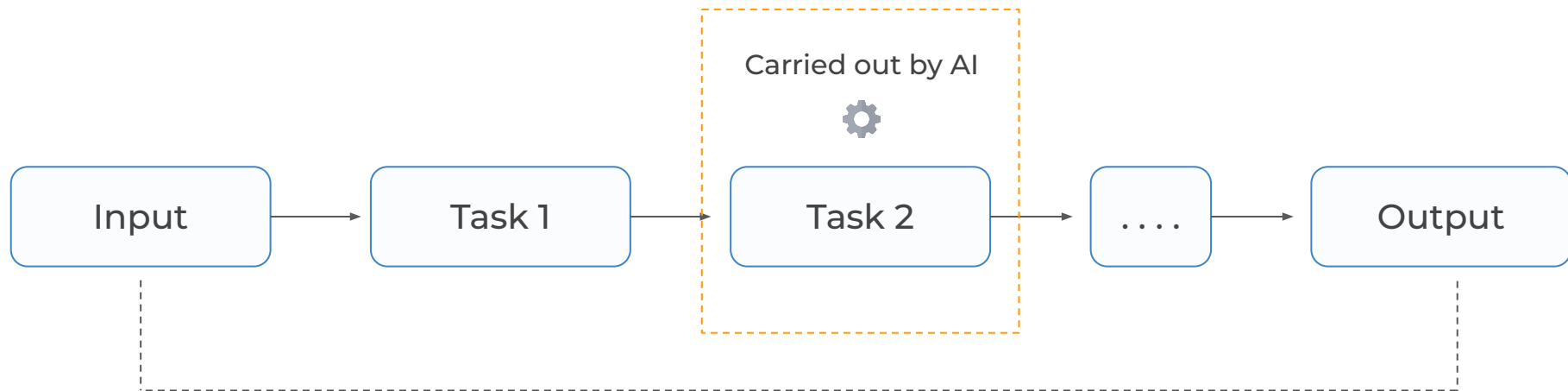
AI workflows are predefined sequences of steps that **use artificial intelligence** to complete tasks automatically



Automated Workflow



AI Automated Workflow

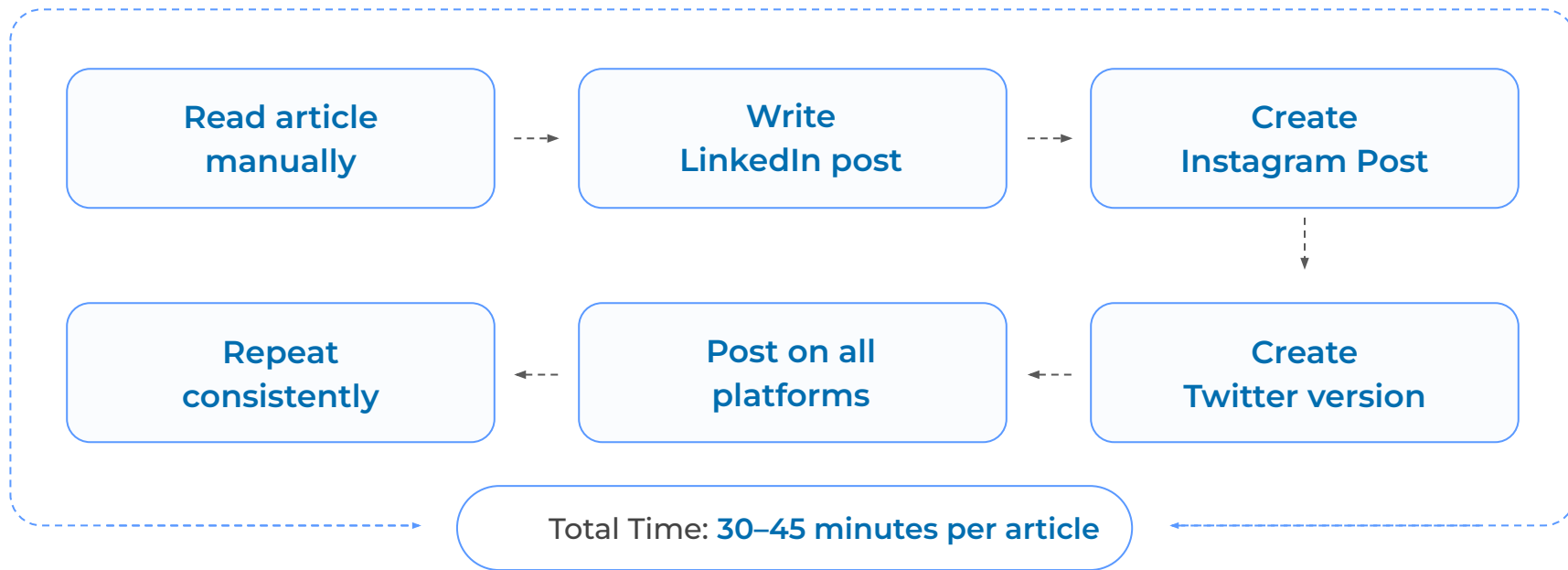


AI Workflows - Example

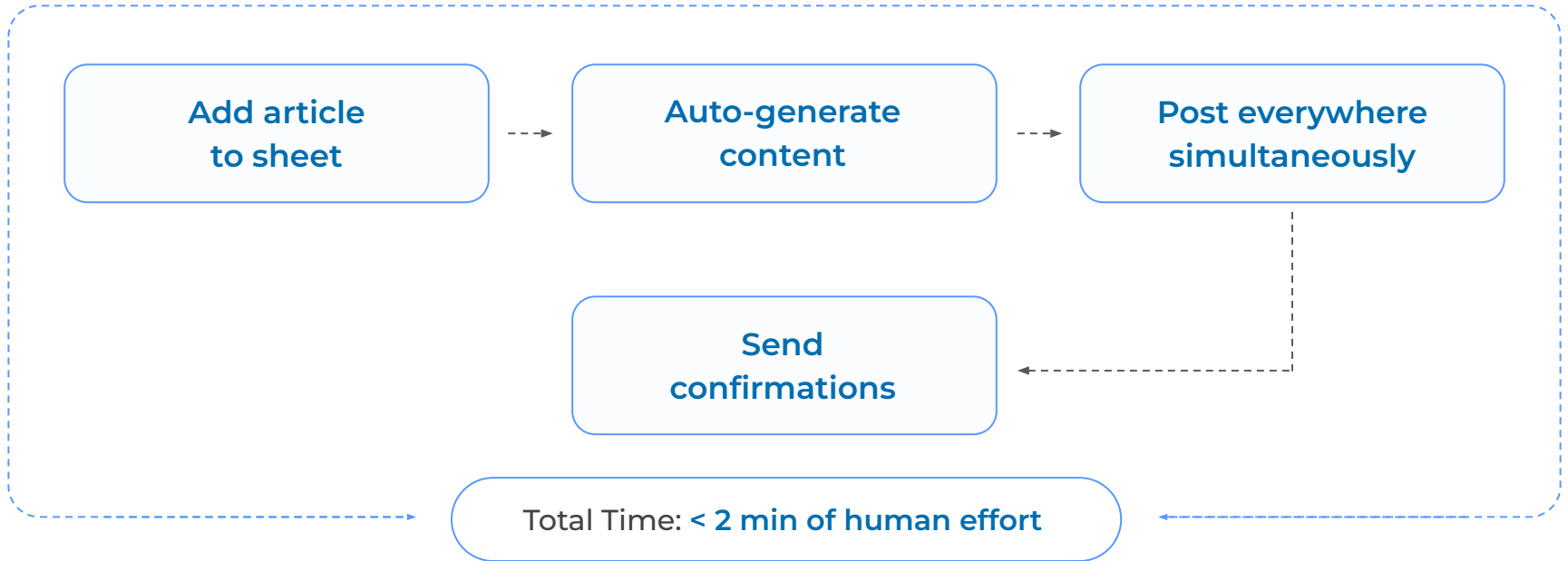
AI-Powered Social Media Content Automation



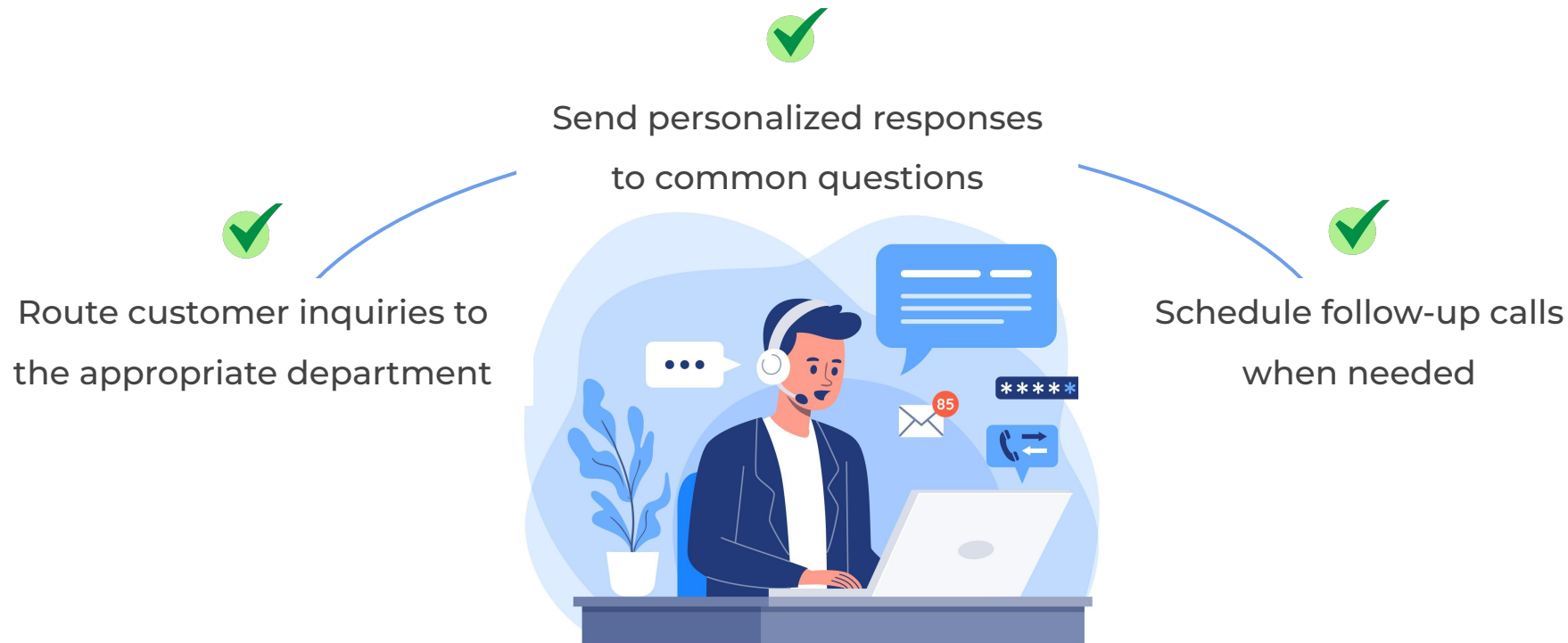
Manual Process



Workflow Process

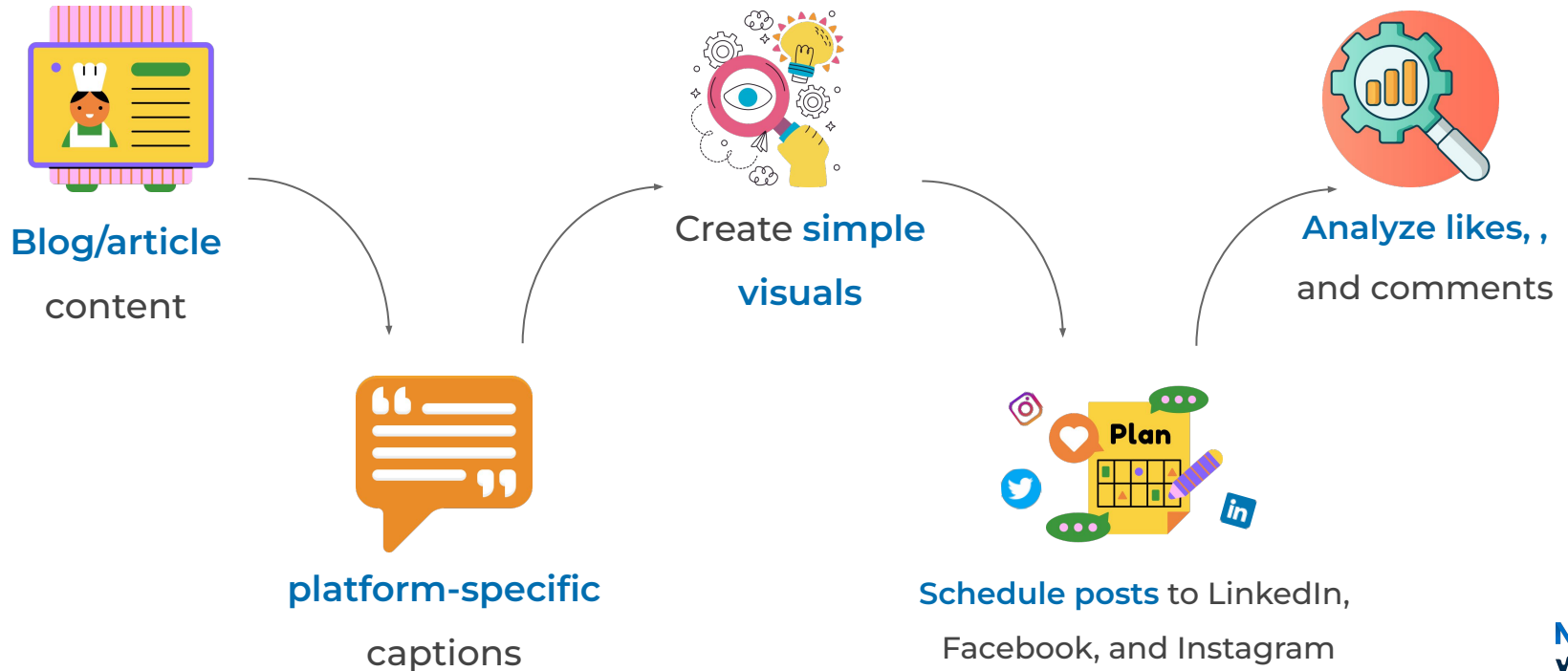


Customer Service

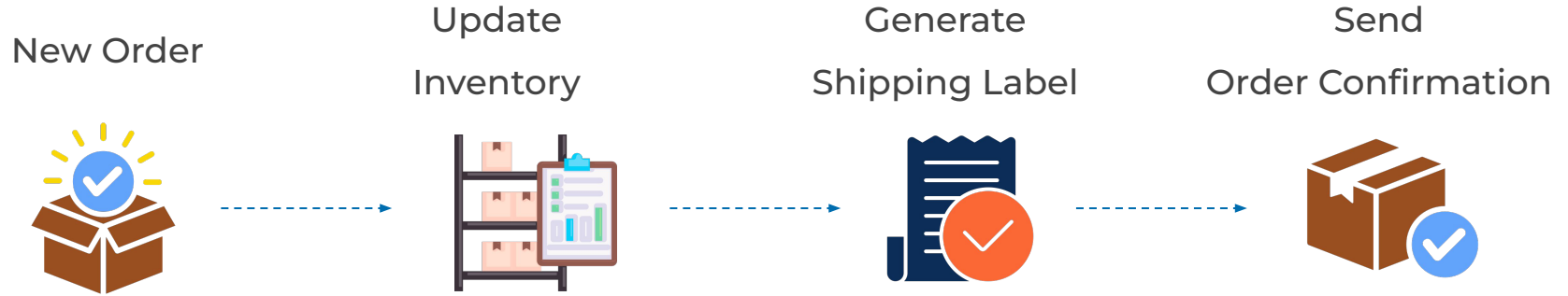


Ex: **Zendesk**, **Atlassian**, **HubSpot**

Social Media Content Creation & Scheduling



E-Commerce Operations



AI Workflows

Predefined Paths

The workflow follows a fixed sequence of steps set by humans

Human Decision-Making

Humans must manually adjust or iterate the workflow if the output is unsatisfactory

Summary

1. AI Workflows
2. Applications of AI Workflows

Quiz Time!

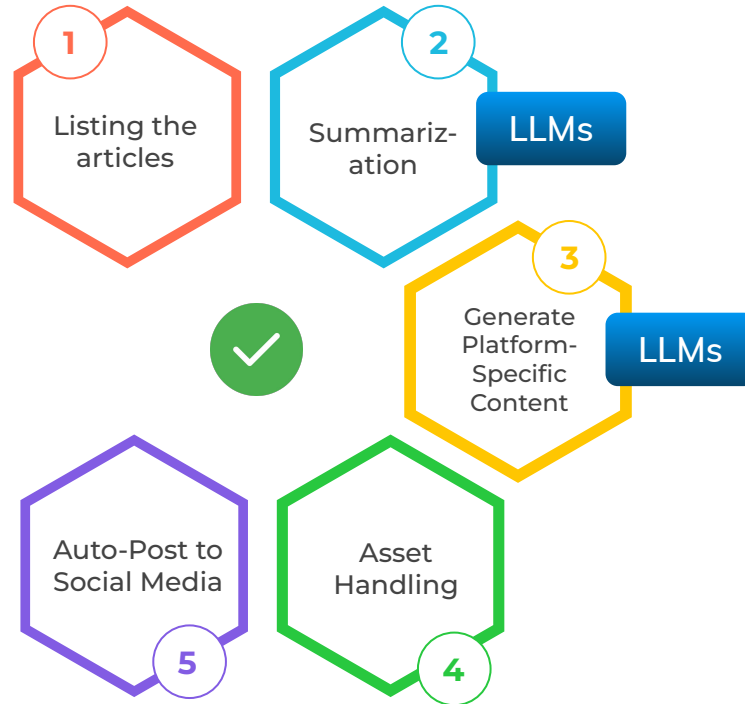


Let's Build AI-Powered Social Media Content Automation Workflow



AI Workflows - Example

AI-Powered Social Media Content Automation



AI Workflows

Building AI Workflows



...many more

Each of these platforms allows us to build AI workflows
using a no-code approach

AI Workflows

Using n8n



...many more

Let's see how can we build AI workflow using **n8n.com**

Using n8n



n8n is a workflow automation platform
that enables users to automate tasks with ease

AI Workflows

Using n8n



It also allows us to connect with popular services like Google Sheets, Slack, Twitter, GitHub, and hundreds more

Building Blocks



Nodes

Each step in the workflow is represented by a **node** that performs specific actions such as retrieving, transforming, or sending data



Building Blocks of n8n

Nodes

Chat Trigger

Initiates workflows
based on user interactions

**Social Media
Nodes**

Manage and interact
with social media platforms

SplitInBatches

Divides a large set of data
into smaller batches,

...many more

Building Blocks of n8n

Nodes

**Google Sheets
Trigger**

Reads from and writes data
to Google Sheets

**Gmail, Slack,
Notion, etc**

Manage and interact
with these platforms

HTTP Request

Used to make API calls
to external services

...many more

Building Blocks of n8n

Connections

A connection establishes a link between nodes to route data through the workflow



Quiz Time!



Creating an AI Workflow for Social Media Content Automation

Pre-requisites

- Google Account
- LinkedIn Account
- X Account (formerly Twitter)



Open n8n

 **AI Tutor** Notes WorkFlow ⌵

Instructions

Go through the below steps to complete this workflow:

- Open the workflow using **Open Workflow** Button
- Try out the workflow you learned during the session.

Open Workflow 



Note

IMPORTANT UPDATE



X (formerly Twitter) Developer Portal has been updated to a **pay-for-use** model. New developer accounts now require a paid plan to access API features, which results in a **"Payment Required" error** during workflow execution



Note

IMPACT 

Posting to X is not possible

with newly created developer accounts



Steps to be Followed

1

Add Google Sheets Trigger

2

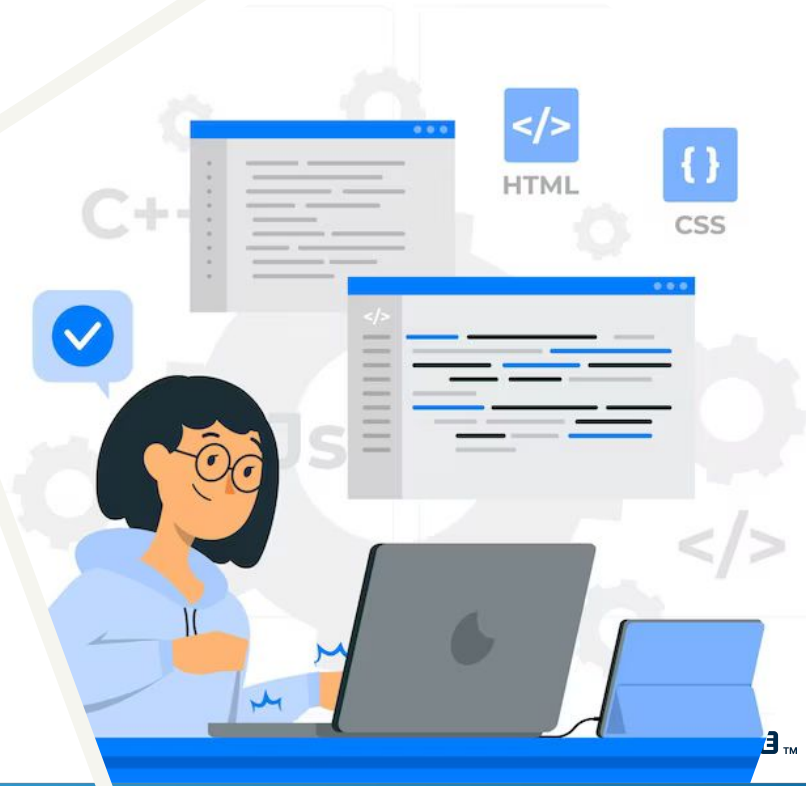
Summarize the Article
Using Gemini model

3

Generate Platform-Specific Content

4

Auto-Post to Social Media



Adding Trigger - Google Sheets

This is the starting point that initiates our workflow

1. Prepare a Google Sheet and add article links
2. Add Google Sheets trigger in n8n
3. Configure the node



Adding Trigger - Google Sheets



Note

Gmail Account: Use gmail.com domain

OAuth Redirect URL:

<https://workflow.ccbp.in/rest/oauth2-credential/callback>

Authorized domains: ccbp.in

Steps to be Followed

1

Add Trigger

2

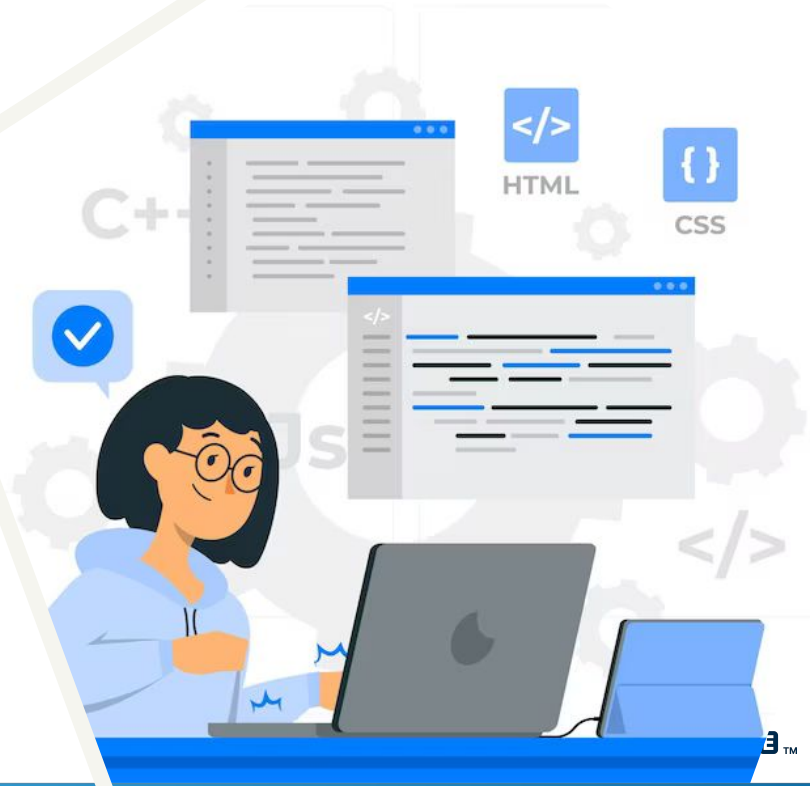
Summarize the Article
Using Gemini

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Auto-Post to Social Media



Summarize the Article Using Gemini

1. Add Basic LLM Chain Node
2. Define a Prompt for Generating Summary
3. Search and Add **Google Gemini Chat Model**
4. Choose `models/gemini-2.5-flash` and Execute the Node

Note

IMPORTANT



As the **models are on the free tier**, we may experience rate limit issues during usage. Use an **alternate API key** for continuation when limits are reached.



Summarize the Article Using Gemini

Advanced Prompt You Can Try

System: "You are a social media strategist with expertise in content adaptation"

User: "Analyze this article and create a summary that:

1. Captures the 3 most important insights
2. Includes any actionable advice or tips
3. Maintains the original tone and perspective
4. Highlights implications for the target audience
5. Stays under 200 words for social media adaptation

Article: {{\$json.text}}



Steps to be Followed

1

Add Trigger

2

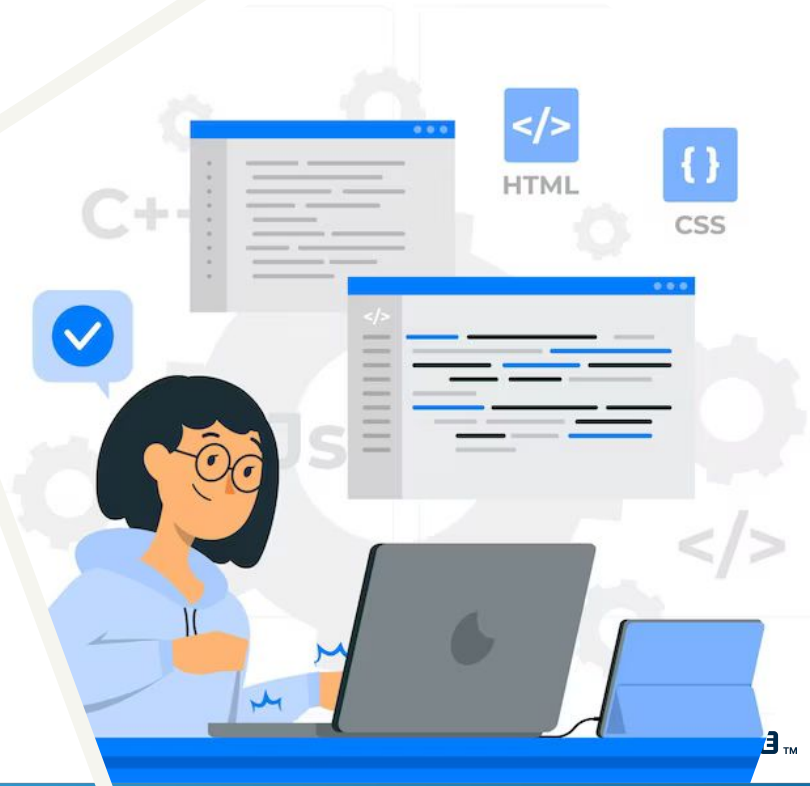
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Why Platform Specific Content?



LinkedIn: Professional, detailed, business-focused



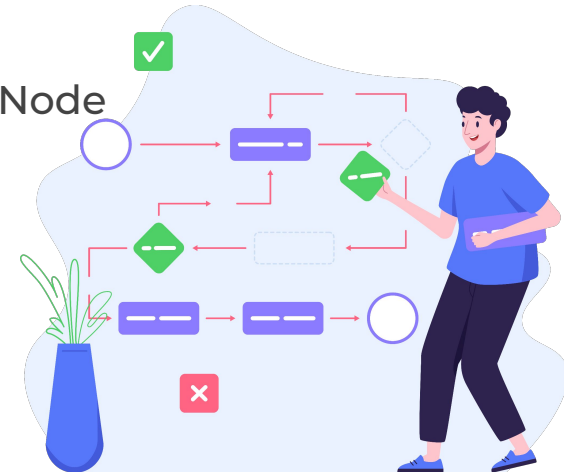
Twitter: Concise, engaging, conversation-starting



Instagram: Visual, inspirational, hashtag-optimized

Generating the LinkedIn Post Content

1. Add Another LLM Chain Node
2. Define Prompt for Creating LinkedIn Post Content
3. Search and Add Google Gemini Chat Model
4. Choose models/gemini-2.5-flash and Execute the Node



Prompt for LinkedIn Post

Assume the role of an industry expert and draft a LinkedIn post discussing key points from a news article relevant to artificial intelligence. The post should provide insightful analysis, connect with current industry trends, and encourage professional engagement or discussion. Highlight any notable implications for the industry. Since LinkedIn supports detailed posts, ensure it is informative and structured. Include a professional call to action. Here's a summary of the article:

{ADD PREVIOUS NODE'S OUTPUT}



Auto-Post to LinkedIn

1. Add LinkedIn Node
2. Select "Create a Post"
3. Authenticate Using Your LinkedIn Credentials
4. Select Username and Fill in the Text to Be Posted
5. Execute the workflow

NOTE

Save and Reload the workflow after connecting to linkedIn

Generating the X(Twitter) Post Content

1. Add Another LLM Chain Node
2. Define Prompt for Creating
X (formerly Twitter) Post Content
3. Search and Add **Google Gemini Chat Model**
4. Choose models/gemini-2.5-flash and
Execute the Node



Prompt for X(Twitter) Post

Draft an X post as an AI industry expert analyzing a key news article. The post should:

Deliver sharp insights on the article's main points

Connect to current AI trends

Spark discussion with a thought-provoking question

Be punchy and engaging

End with a clear call to action

Length: Within 30 words

Focus: What matters most to the AI community. Make every word count.

Article summary:

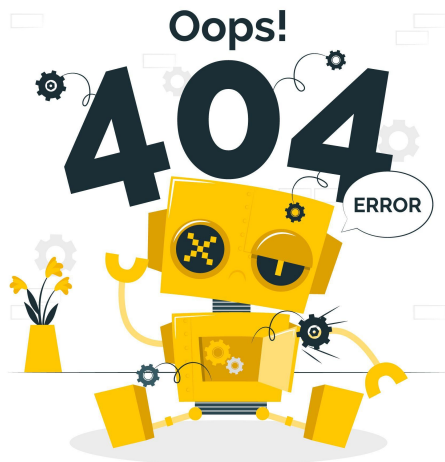


Warning

Don't Make Post Too Lengthy

You may sometimes encounter
a 402 error if your post is too long

Make sure the length doesn't exceed
the allowed limit



Steps to be Followed

1

Add Trigger

2

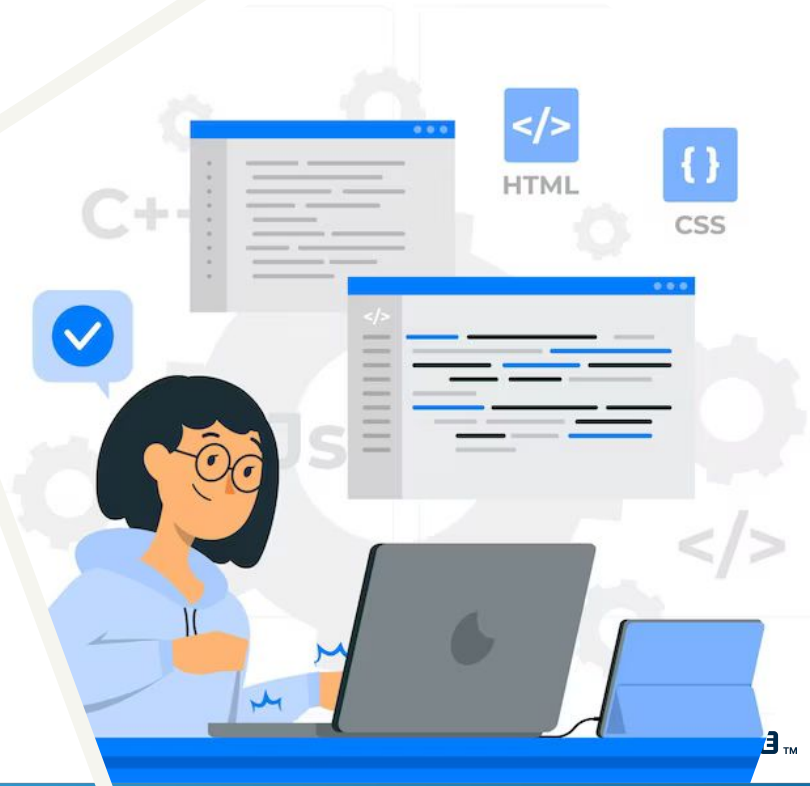
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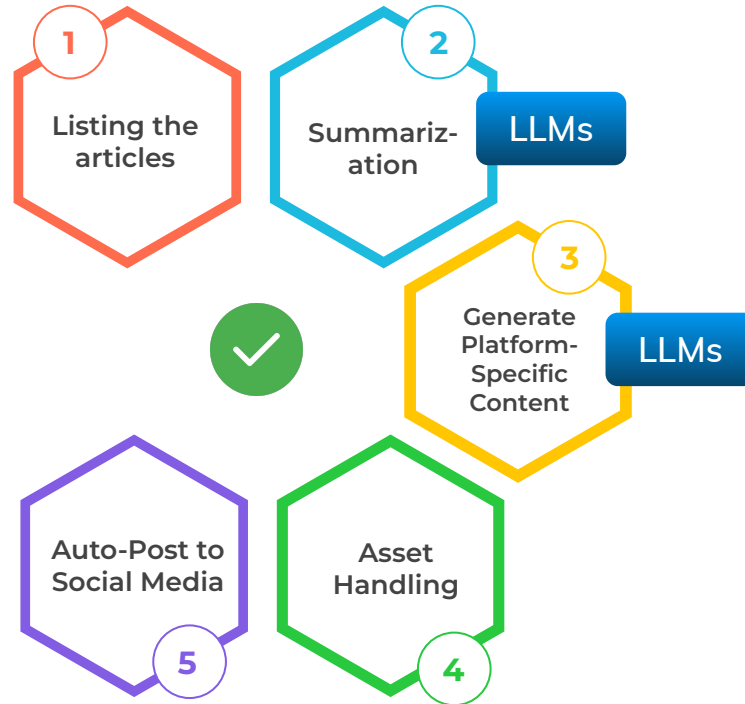
Auto-Post to X(Twitter)

1. Add X (Twitter) Node
2. Authenticate to Your X Account
3. Create Your X API Credentials for Client ID and Secret Key
4. Fill in the Text to Be Posted



AI Workflows - Example

AI-Powered Social Media Content Automation



Workflow Testing

➤ Trigger Testing

- Add test article to Google Sheet
- Execute the workflow
- Check data passes between nodes

Workflow Testing

- **Content Quality Testing**
 - Review AI-generated summaries
 - Verify platform-specific adaptations
 - Check character limits and formatting

Workflow Testing

- **End-to-End Testing**
 - Complete workflow execution
 - Verify posts appear on platforms
 - Confirm success notifications

Common Issues you may Encounter



402 Errors

Content length exceeds the maximum characters allowed



Authentication Failures

API credentials expired or incorrect

Common Issues you may Encounter



Rate Limiting

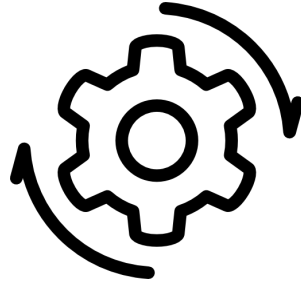
Too many API calls in
short timeframe



Content Quality

Prompts need refinement
for better output

Setting the Scheduler to Run Every Week



Common Scheduling Options

1

Manual Trigger

Run on-demand
for testing

2

Time-based Schedule

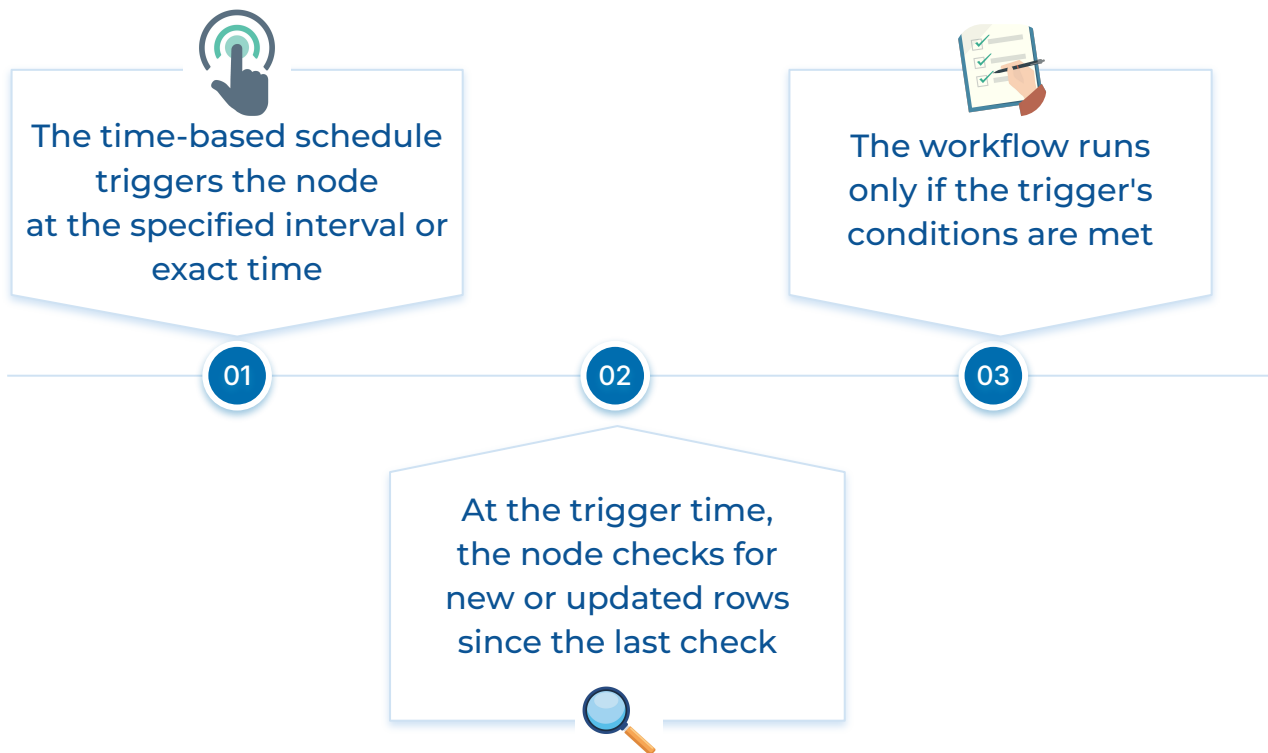
Every 15 minutes,
hourly, daily

3

Event-based

Triggered by external
webhooks

Time Based Schedule



Setting the Scheduler in n8n

Time Based Schedule

n8n triggers the workflow only for new or updated rows since the last check, not for every row in the sheet



n8n

Automating the Execution

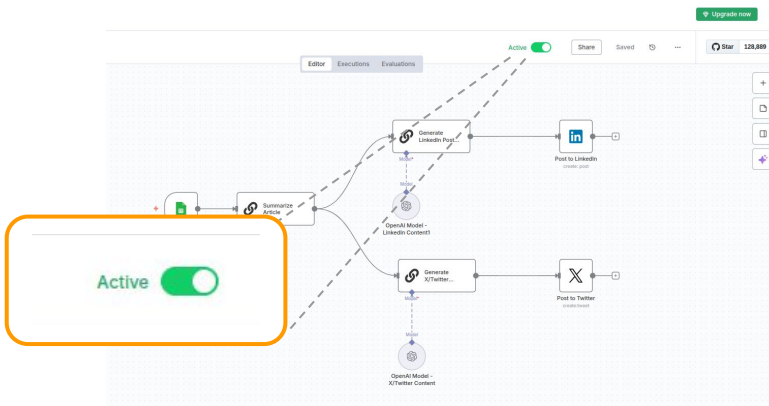


Try automating the execution of our workflow every week at a specific time

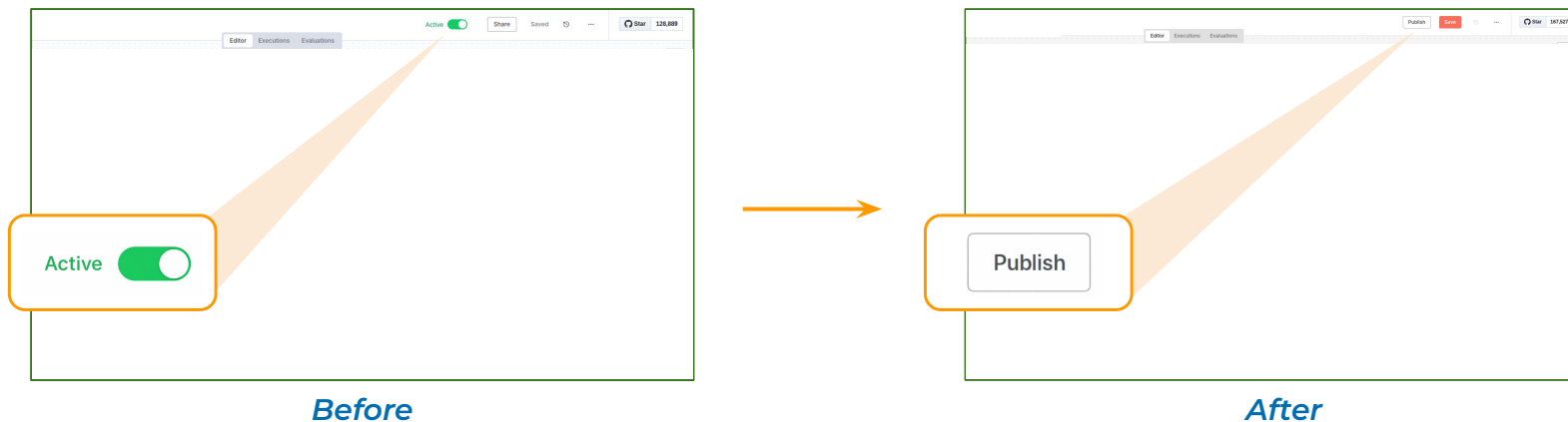
Automation with n8n

Workflow Should be in Active State

To ensure your workflow is automated and triggers as expected, make sure the workflow is active



Button Updation by n8n



The “**Active**” button has been updated to “**Publish**”, allowing you to publish your workflows. Kindly note that the Publish feature is restricted to a limited number of uses on the portal

Key Takeaways

AI Workflows

- ✓ Applications
- ✓ Tools for building AI Workflows

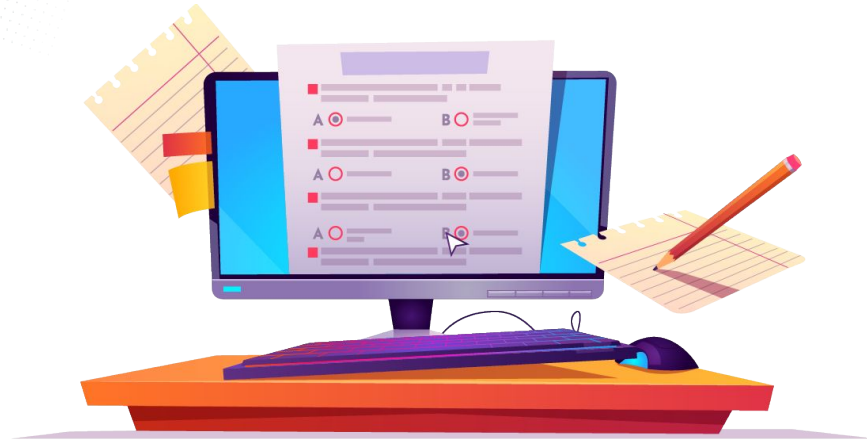
Creating an AI Workflow

Creating an AI Workflow
Using **n8n**

Quiz Time!



Practice



Feedback





THANK YOU





ALL THE BEST



Creating an AI Workflow

Step 1

Sign up and create a new scenario.



Create a free account
on **make.com**



Click
"**Create a new scenario**"



You'll see a **blank canvas**
where your workflow is built

Creating an AI Workflow

Step 2

Adding Trigger.

The trigger is **what starts your workflow**

workflow



Trigger

Creating an AI Workflow

Step 2

Adding Trigger.

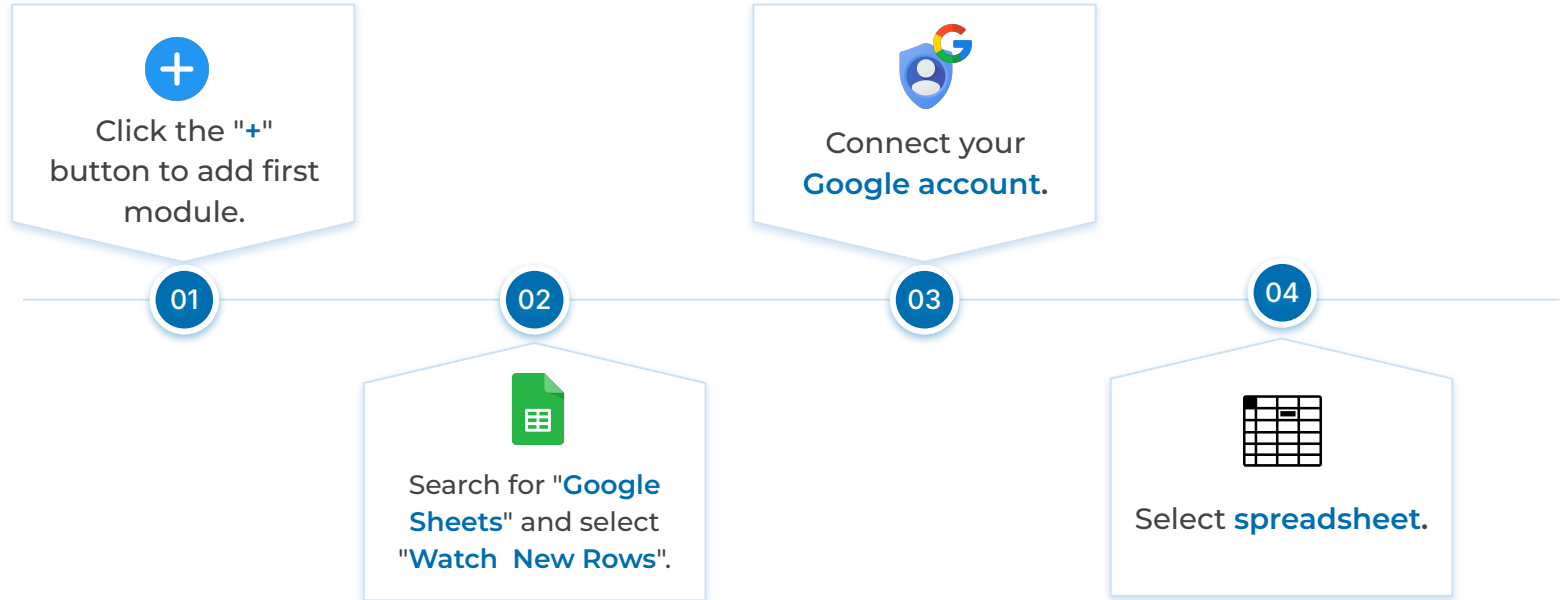
Here, let's add all of our articles in the google sheets
LLM should read this articles and generate summary

What it does:

- ★ Monitors a sheet for new rows.
- ★ Each row contains an article (either link or raw text).
- ★ Triggers the automation when a new article is added.

Adding a Trigger

Step 2



Article Summarization and Social Media Posting

Adding a Trigger



Adding a Trigger

IMPORTANT



If you manually paste data into an old row
or modify an existing row: It will not trigger the
workflow unless: You reset the starting point using

“Choose where to start” in [Make.com](https://www.make.com).



Steps to be Followed

1

Create a scenario

2

Add Trigger

3

Summarize the Article Using AI

4

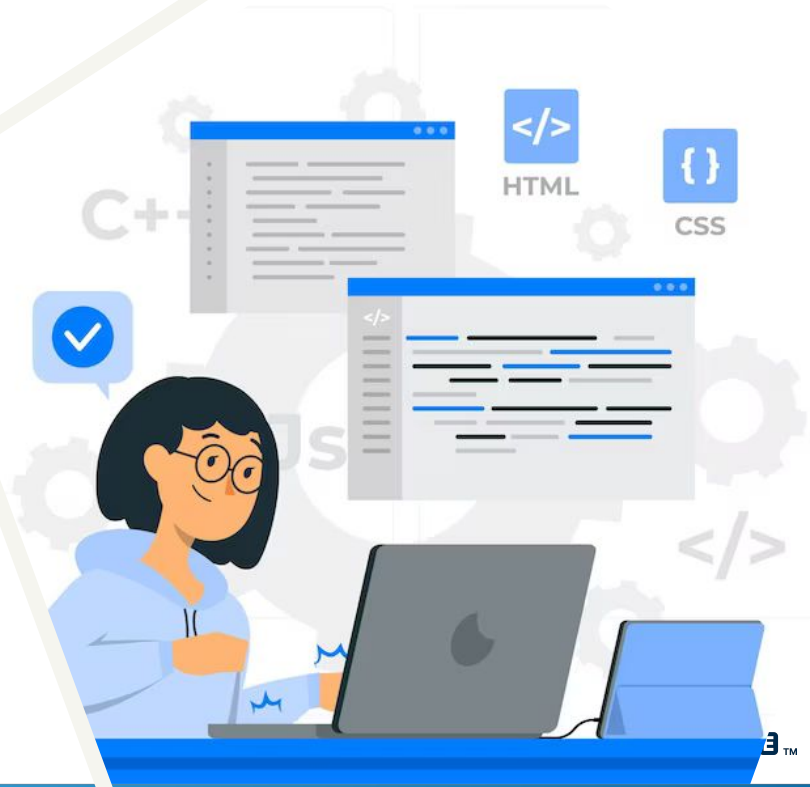
Route to Different Social Platforms

5

Generate Platform-Specific Content

6

Auto Post to Social Media



Creating an AI Workflow

Step 3

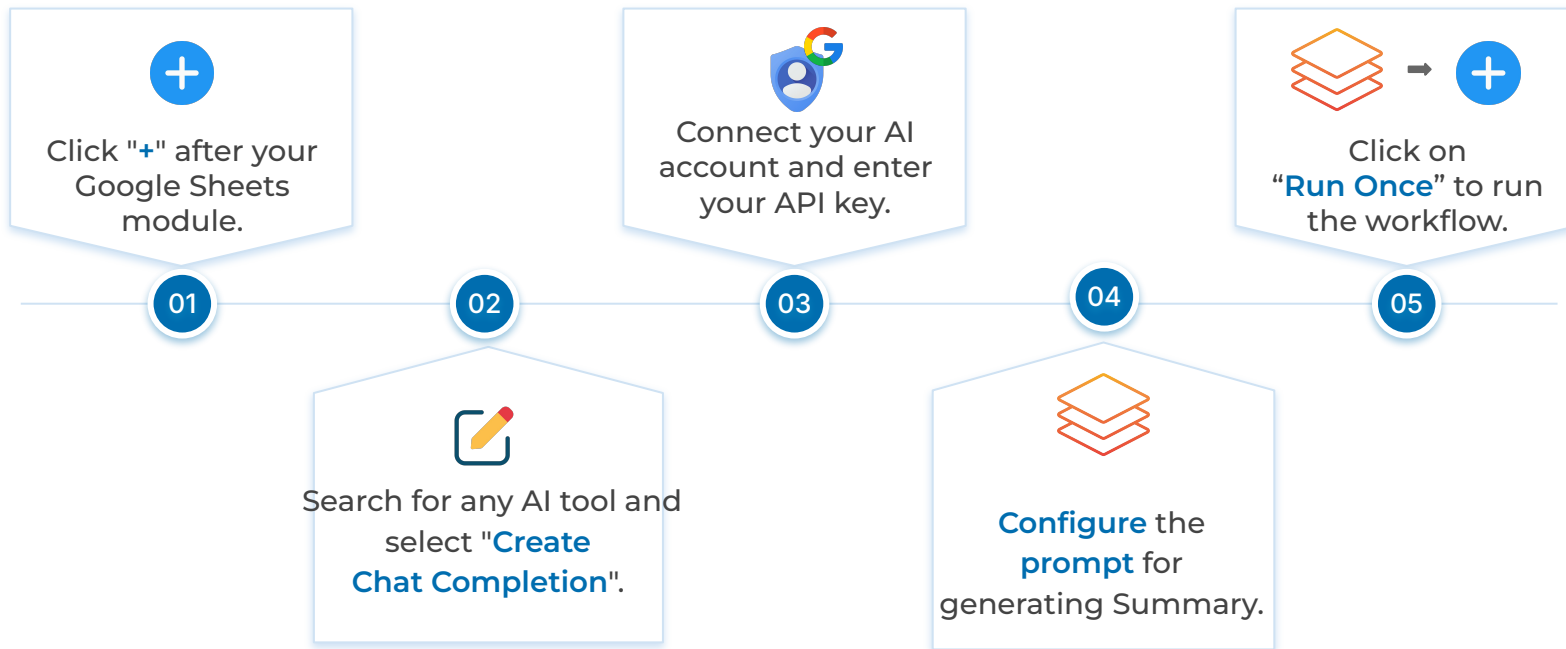
Summarize the Article Using AI.

- ★ The article will be **sent to an AI module** (Groq) to summarize the content.
- ★ The **AI will process the article and return a brief**, concise summary, making it suitable for sharing on social media.

Popular LLM Examples

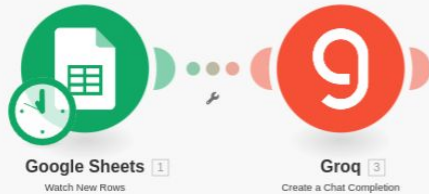


Summarizing the Article Using AI



Creating an AI Workflow

Summarizing the Article Using AI



Steps to be Followed

1

Create a scenario

2

Add Trigger

3

Summarize the Article Using AI

4

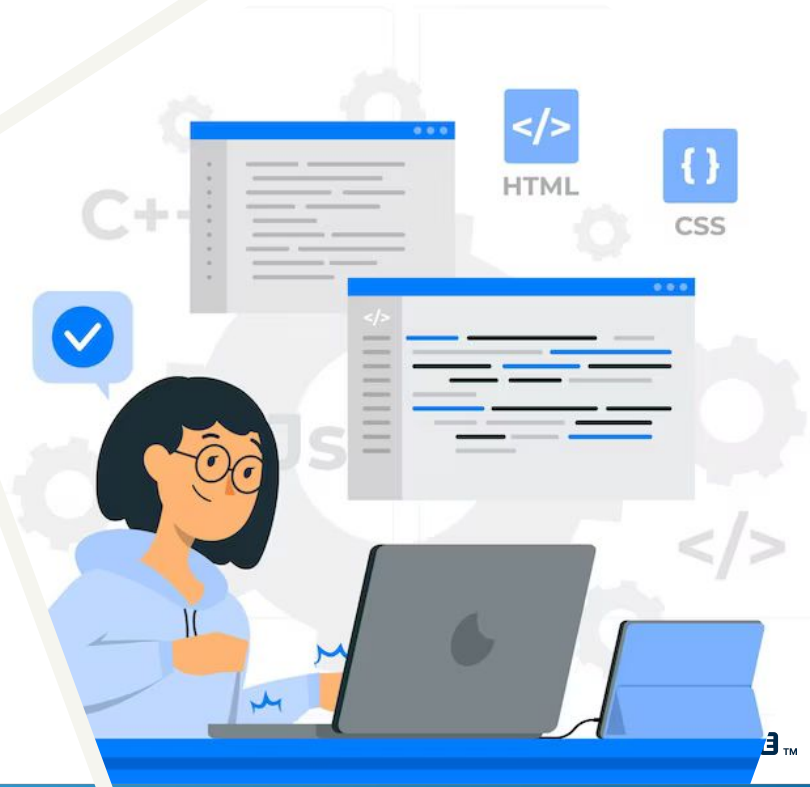
Route to Different Social Platforms

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Auto Post to Social Media



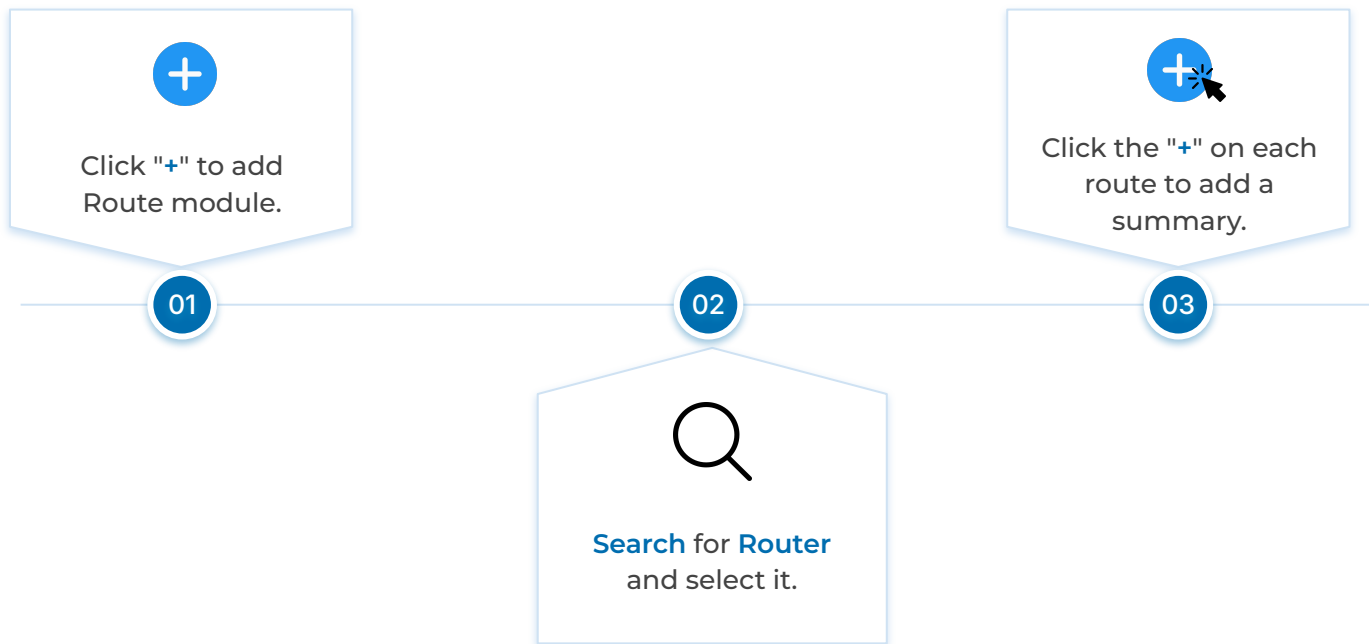
Creating an AI Workflow

Step 4

Adding Router to Different Social Platforms

- Using a router module, we will split the workflow into multiple paths:
LinkedIn, Facebook, Instagram
- Each platform should get a **customized version of the post**, based on its specific format and tone

Adding Router to Different Social Platforms



Steps to be Followed

1

Create a scenario

2

Add Trigger

3

Summarize the Article Using AI

4

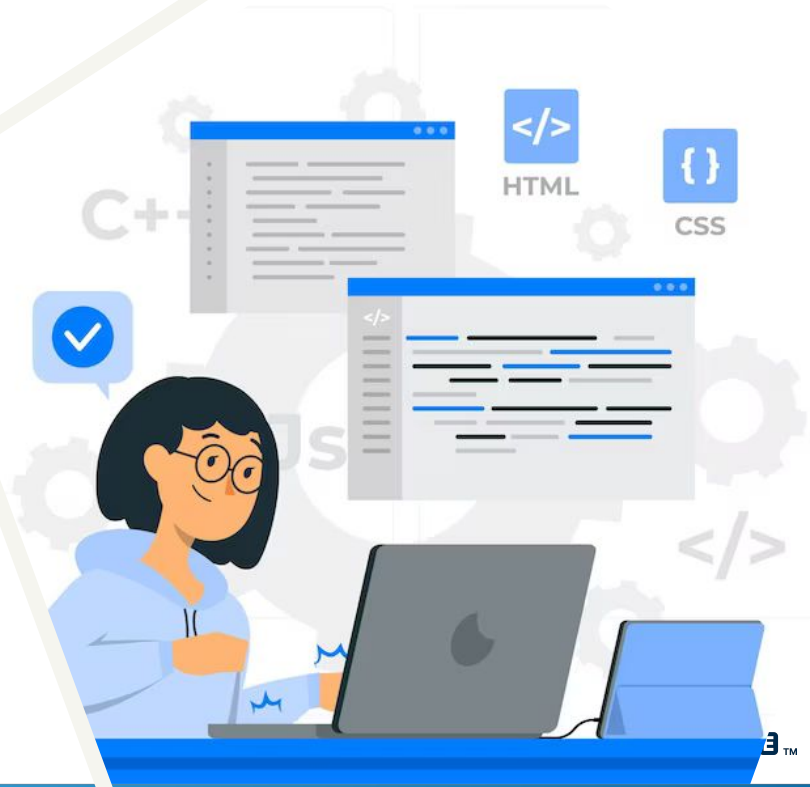
Route to Different Social Platforms

5

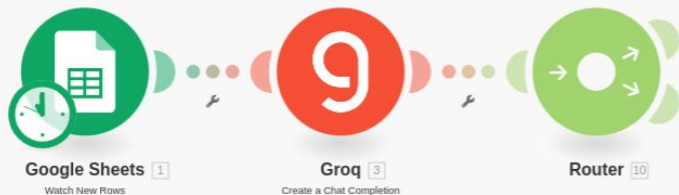
Generate Platform-Specific Content

6

Auto Post to Social Media



Adding Router for Different Platforms



Creating an AI Workflow

Step 5

Generate Platform-Specific Content

For each platform, create a specific prompt, like:

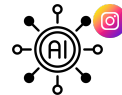
1. For **LinkedIn**: Professional summary with business insights
2. For **Facebook**: More conversational content with call to action
3. For **Instagram**: Visual-focused description with hashtags

Generate Platform-Specific Content



Select any **AI Tool** and
add prompt for
LinkedIn post.

01



Select any **AI Tool** and
add prompt for
Instagram post.

03



Select any **AI Tool**
and add prompt for
Facebook post.

02

Generate Platform-Specific Content



Steps to be Followed

1

Create a scenario

2

Add Trigger

3

Summarize the Article Using AI

4

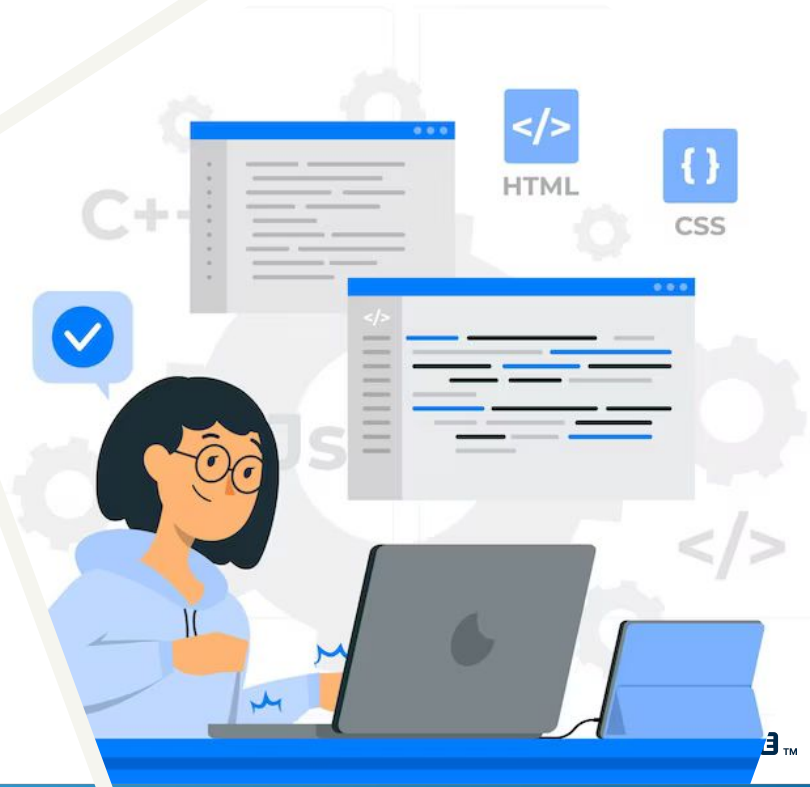
Route to Different Social Platforms

5

Generate Platform-Specific Content

6

Auto Post to Social Media



Creating an AI Workflow



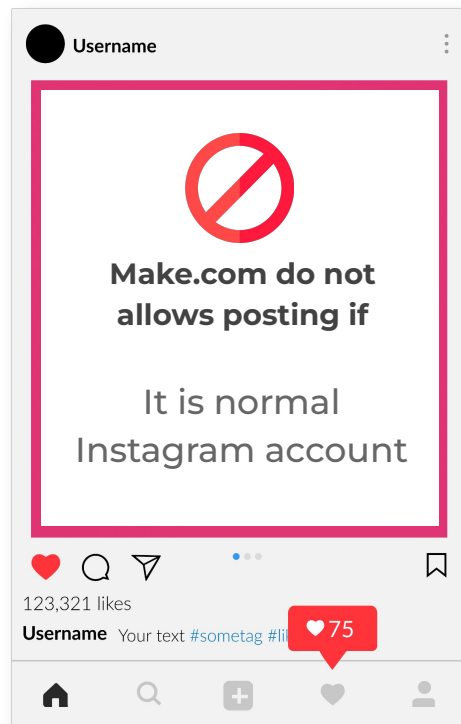
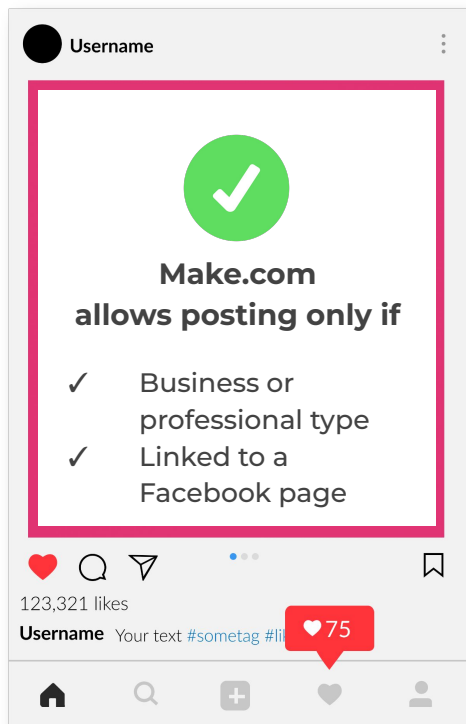
Step 6

Auto Posting to Social Media

Finally, we will **use platform-specific modules to automatically post the generated content** to the selected social media platforms.

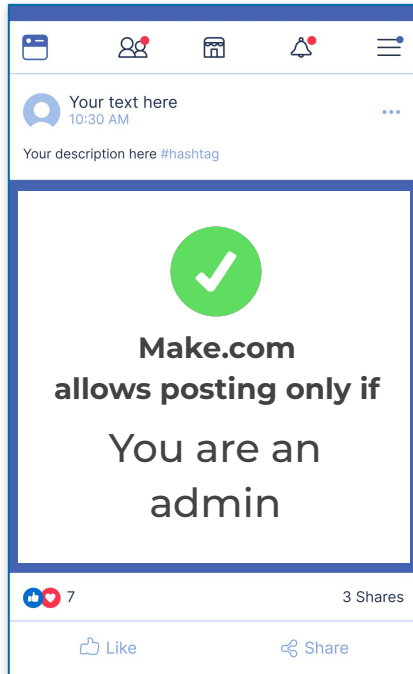
Make.com

Note for Instagram Post



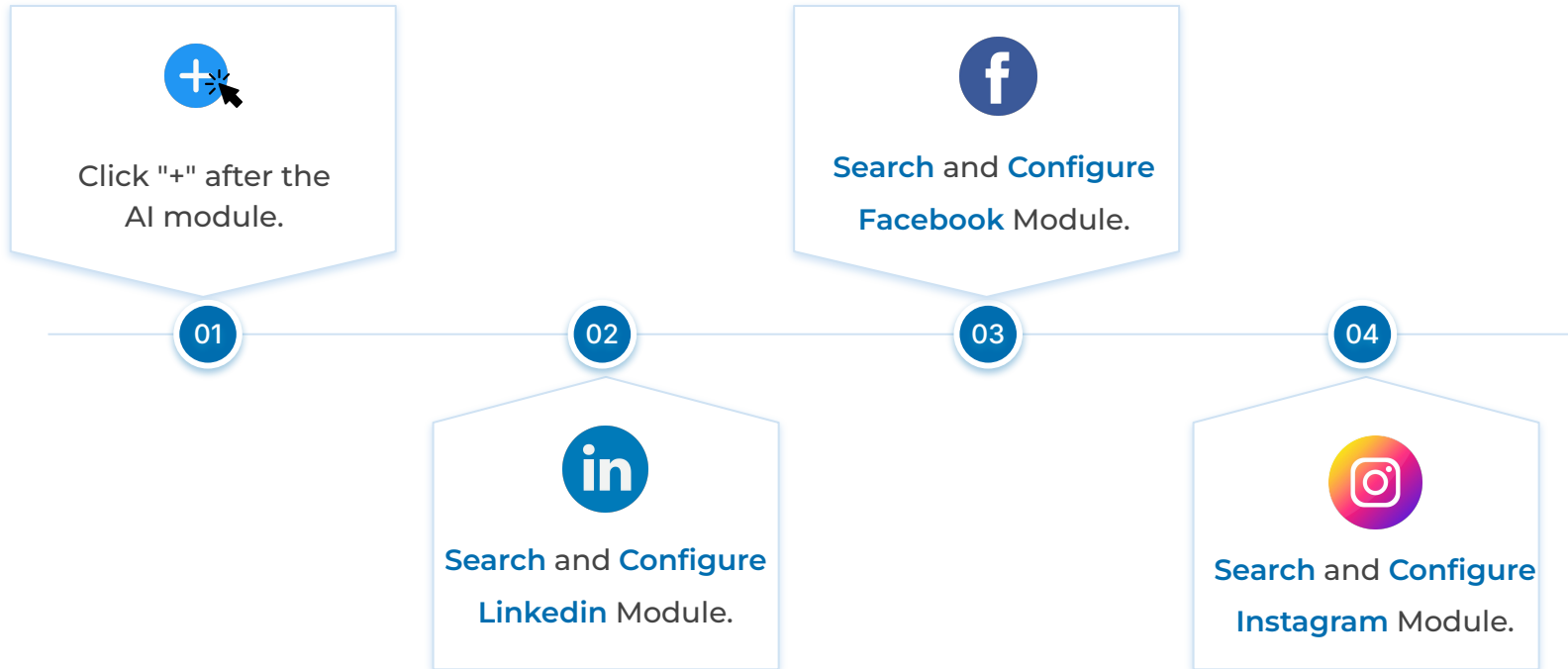
Make.com

Note for Facebook Post



Make.com **allow you to publish, update, or monitor posts, photos, and videos** only on Facebook Pages where **you are an admin.**

Auto Posting to Social Media

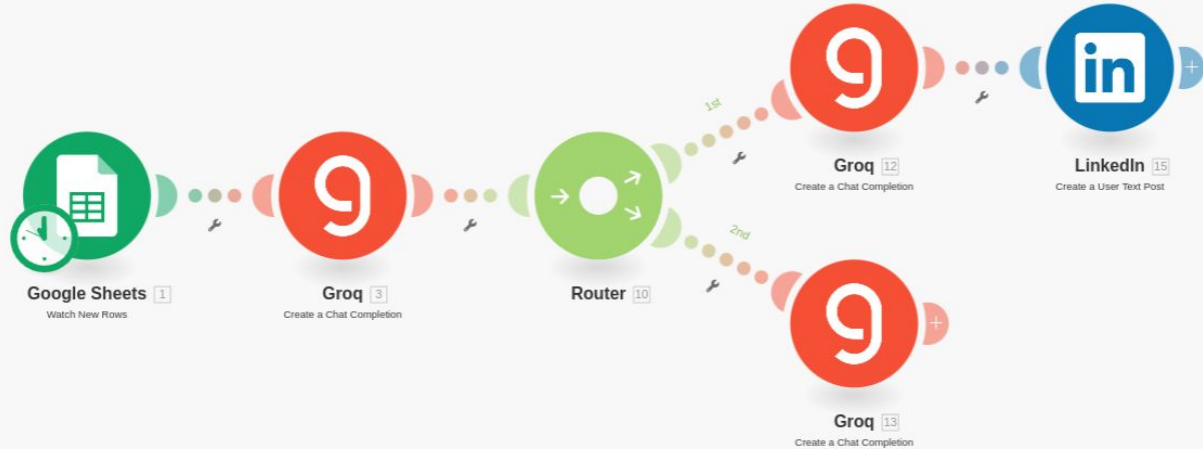


Prompt for LinkedIn Post

Assume the role of an Industry expert and draft a LinkedIn post discussing key points from a news article relevant to artificial intelligence. The post should provide insightful analysis, connect with current industry trends, and encourage professional engagement or discussions. Highlight any notable implications for the industry. LinkedIn supports detailed posts, so ensure it is informative and structured. Add a professional call to action. Here's a summary of the article:



Generate Platform-Specific Content



Prompt for Facebook Post

Act as a social media manager and write a Facebook post. The post should be concise, impactful, and highlight an important aspect or benefit. Include an element of curiosity. Keep within Facebook's 280-character limit. Use relevant hashtags for greater visibility: [#Hashtag1, #Hashtag2].

Here's a summary of the article that I want you to turn into a Facebook post:



Prompt for Instagram Post

As a digital marketing specialist, create a n Instagram post about the following artificial intelligence article. Ensure the content is visually appealing and includes an inspirational message. Use emojis to enhance engagement. Suggest relevant hashtags: [#Hashtag1, #Hashtag2].

Here's a summary of the article that I want you to turn into an instagram post:



Example

Prompt

By how many runs did England win the world cup final?

Generate ↑

England won the world cup final in 2019 by defeating New Zealand. England was declared the winner based on the number of boundaries scored throughout the match. England scored 26 boundaries (4s and 6s) in total, while New Zealand scored 17 boundaries

Example

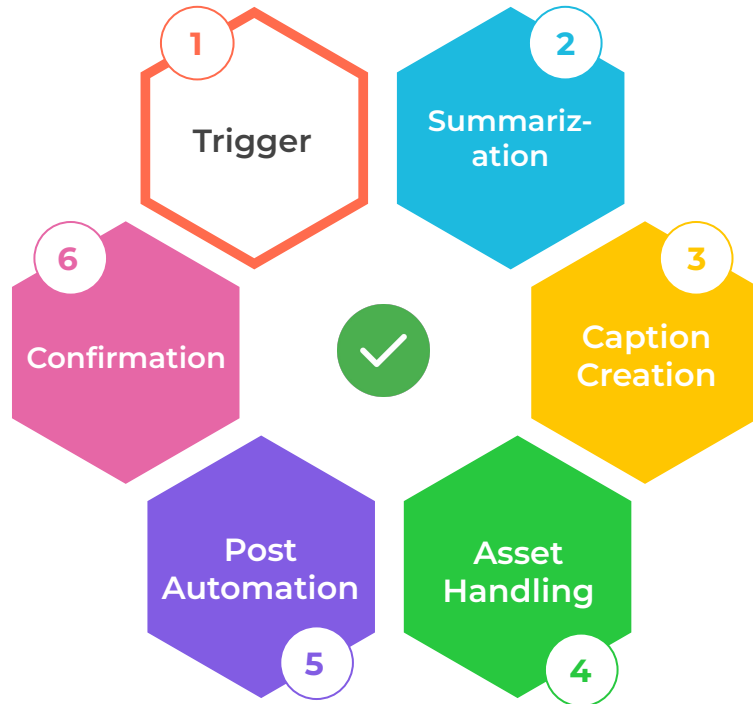
Prompt

Describe the city of Hyderabad in two lines

Generate ↑

Hyderabad, known as the "**City of Pearls**," seamlessly blends its rich historical heritage with a vibrant, modern urban landscape, offering a taste of both the past and the present

Step-by-Step Workflow



Trigger :

You upload or paste the article.



Step-by-Step Workflow

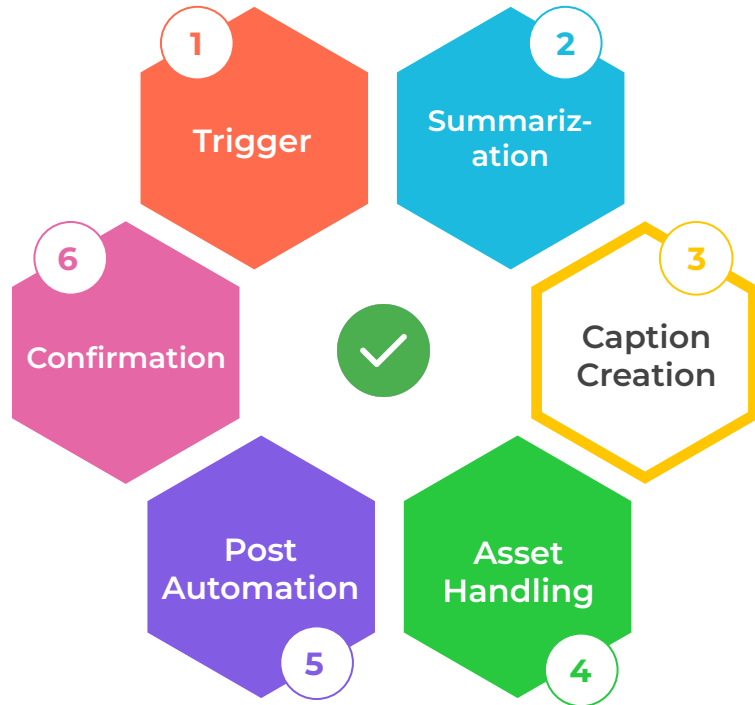


Summarization :

LLM reads and summarizes the article.



Step-by-Step Workflow

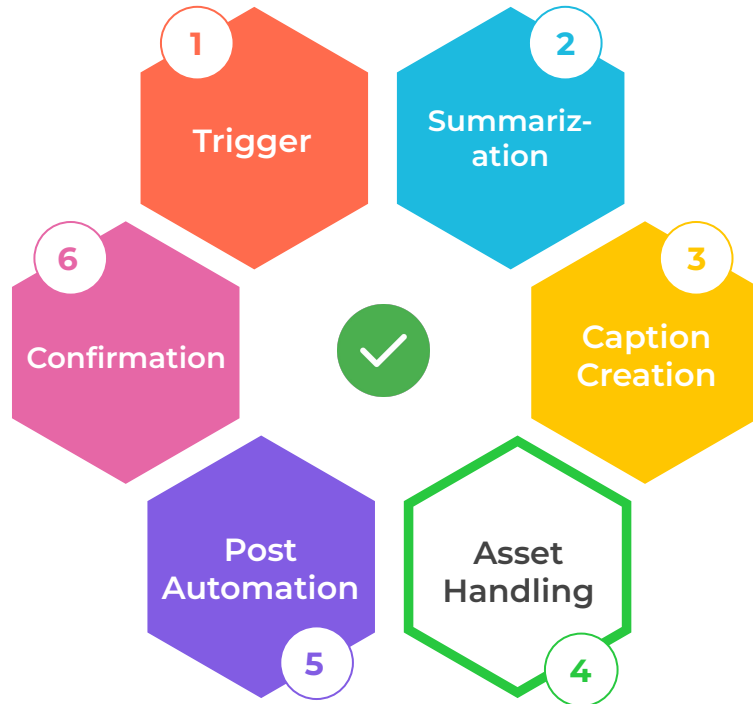


Caption Creation :

LLM creates
platform-specific
captions with hashtags.



Step-by-Step Workflow

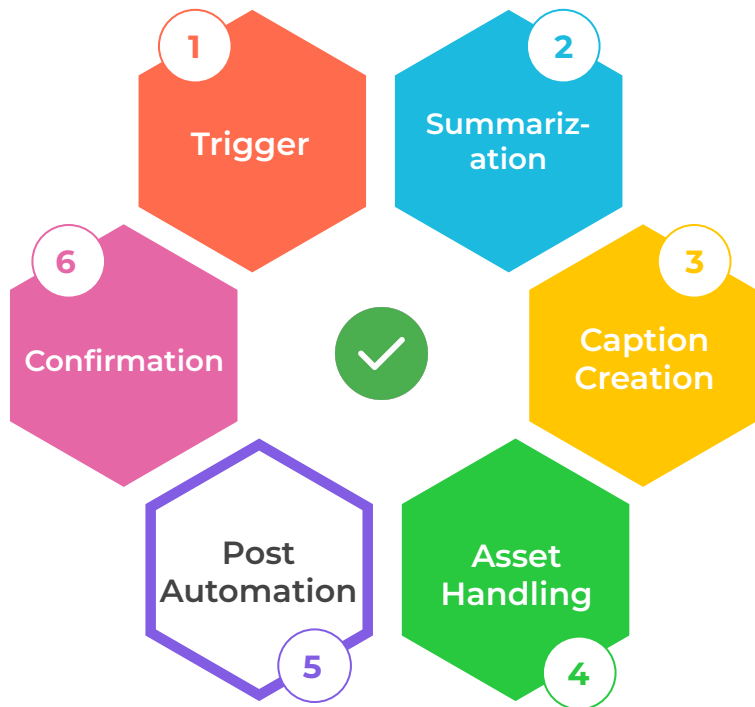


Asset Handling :

Auto-generates an image
via Canva or DALL·E



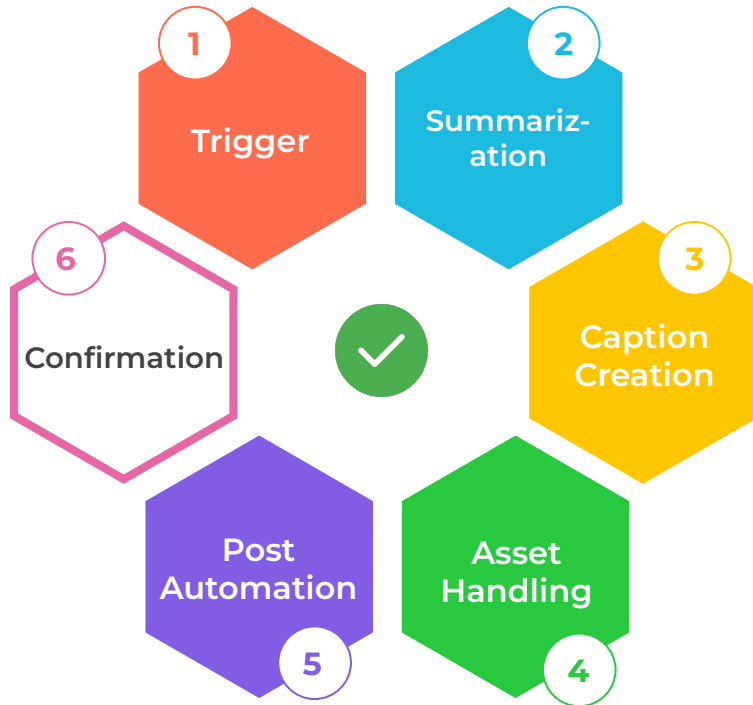
Step-by-Step Workflow



Post Automation :

- Uses [LinkedIn API](#) to post to your profile or page.
- Uses [Instagram API](#) to upload a carousel/story.
- Uses [X \(Facebook\) API](#) to tweet it.

Step-by-Step Workflow



Confirmation :

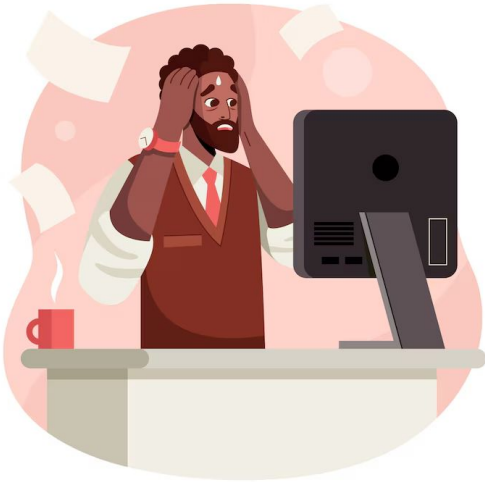
Sends you a message like

" Posted on all 3 platforms!"

Summarizing an Article and Posting



Social Content Creation & Scheduling



Marketing teams **often spend hours creating, scheduling, and analyzing social posts.**

Creating an AI Workflow

Step 2

Adding trigger - Google Sheets

1. Click the "+" button to add your first module.
2. Search for "Google Sheets" and select "Watch Rows".
 - The "Watch Rows" module checks for new rows added to your selected Google Sheet.
3. Connect your Google account when prompted.
4. Select the spreadsheet and worksheet where articles will be pasted.

AI Workflows

AI Workflows are **predefined sequences of steps that combine AI with other digital tools** to accomplish specific tasks

Think of it like creating a recipe for the AI to follow:

1. You decide all the ingredients and steps in advance
2. The AI executes the predefined steps precisely when needed

Customer Service



Creating an AI Workflow

Step 3

Add your trigger - Google Sheets

4. Configure the **prompt**:

Create a concise summary of the following article in 200 words or less.
Include the main points while maintaining the article's key message.
Article: {{1.column_with_article_text}}

4. Click on “**Run Once**” to run the workflow

What you will learn?

Investments,
Applications & Demonstrations

make.com

Generative AI
Landscape

Understanding LLMs

AI Workflows

Creating an AI Workflow

Step 6

Configure LinkedIn, Instagram, Facebook Publishing Modules

Configure a prompt: **Instagram**

As a digital marketing specialist, create a n Instagram post about the following artificial intelligence article. Ensure the content is visually appealing and includes an inspirational message. Use emojis to enhance engagement. Suggest relevant hashtags: [#Hashtag1, #Hashtag2]. Here's a summary of the article that I want you to turn into an instagram post:

Creating an AI Workflow

Step 6

Configure LinkedIn, Instagram, Facebook Publishing Modules

Configure a prompt: **Facebook**

Act as a social media manager and write a Facebook post about artificial intelligence. The tweet should be concise, impactful, and highlight an important aspect or benefit. Include an element of curiosity. Keep within Facebook's 280-character limit. Use relevant hashtags for greater visibility: [# Hashtag1, #Hashtag2].

Here's a summary of the article that Iwa nt you to turn into a Facebook post:

Setting the Scheduler

1. In the scenario editor, click on the "**Every 15 minutes**" option in the toolbar.
2. The **Schedule settings** panel will appear.
3. Ensure the "**At regular intervals**" option is selected.
4. Set the interval to **15 minutes**.
5. Click "**Save**" to apply the changes

Next Session

Have Limited Knowledge

Despite being trained on vast amounts of data, LLMs have **limited knowledge of proprietary information.**

Ex: personal information or internal company data.



Tech Giants Make Massive Investments into Generative AI

Microsoft adds \$10 billion to investment in ChatGPT maker OpenAI

Microsoft, which plowed \$1 billion into OpenAI in 2019, is seeking an inside edge on some of the most popular and advanced artificial intelligence systems



ChatGPT has lit up the internet since its introduction in late November, gathering a million users in less than a week (Bloomberg)

Microsoft Corp. is making a \$10 billion investment over several years in OpenAI, the pioneering artificial intelligence research lab behind ChatGPT and DALL-E.

Elon Musk Invested \$50 Million in OpenAI: 2 Glaring Concerns He Has Now

In a recent interview on *CNBC*, Elon Musk, **Tesla** founder, CEO, and billionaire entrepreneur, said he invested \$50 million into OpenAI and is ultimately the reason that the company exists.

If you haven't heard of OpenAI, you may have heard of the company's flagship product, ChatGPT, which is a generative *artificial intelligence* tool that can create a wide range of content, including conversations that are strikingly accurate and resemble normal human behavior.

Musk alleges that he came up with OpenAI's name and recruited key engineers but ultimately ended up leaving the company in 2018 due to a conflict of interest with his work at Tesla. There have also been reports that Musk didn't like the direction the company was going in. Here are two glaring concerns Musk has now with OpenAI.

Introduction

AI Workflows

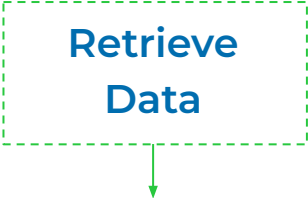
AI Workflows are **predefined sequences of steps** that combine AI with other digital **tools** to accomplish specific tasks.



Basic Building Blocks - Modules

Modules can **perform various tasks**, such as **retrieving data**, **transforming information**, or **executing actions**.

**Retrieve
Data**



Eg: Databases etc.

**Transform
Data**



**Execute
Actions**



Eg: Send Emails / Updates etc.

Basic Building Blocks - Modules

Modules are categorized based on their roles within a scenario.



Triggers

Receiving a new email



Actions

Creating, updating, or deleting data



Searches

Retrieve specific data



Tools

Handle data manipulation



Universal Modules

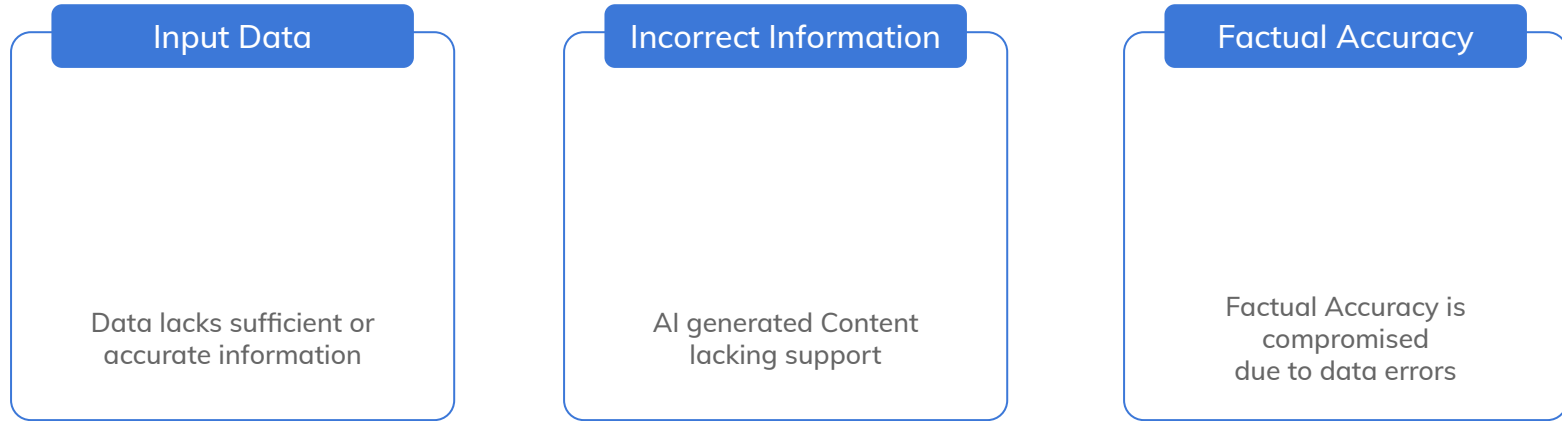
Allow custom API calls

Scenario

A scenario is a **complete automation workflow composed of interconnected modules**



Hallucination



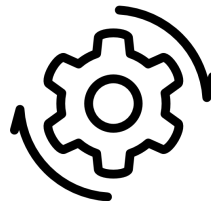
AI can generate incorrect or nonsensical information that isn't supported by the input data or factual accuracy

Important Considerations

1. **Polling Trigger:** The "**Watch Rows**" module in Google Sheets is a polling trigger, meaning it checks for new data at the intervals you've set. It doesn't respond instantly to changes in the sheet.
2. **Manual Execution:** If you want to test your scenario or run it immediately, you can use the "**Run once**" button. However, this won't affect the scheduled execution; it's only for manual testing.

Scenarios

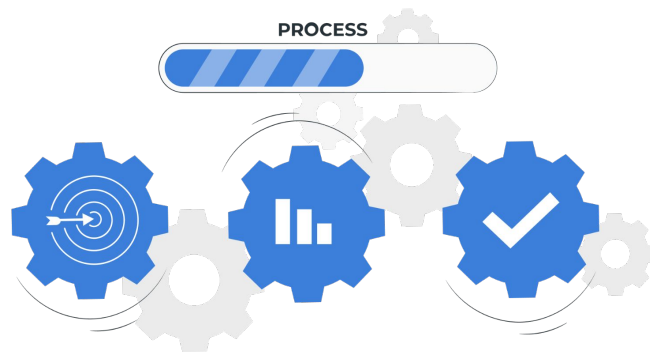
By default, **Make.com scenarios** are
set to run on demand or
immediately.



Building AI workflows

AI Workflows

LLMs **only execute** those **predefined actions**, without trying to decide what should happen next



Prompt for Instagram Post

As a digital marketing specialist, create a n Instagram post about the following artificial intelligence article. Ensure the content is visually appealing and includes an inspirational message. Use emojis to enhance engagement. Suggest relevant hashtags: [#Hashtag1, #Hashtag2].

Here's a summary of the article that I want you to turn into an instagram post:



Introduction

AI Workflows

- ✓ Applications
- ✓ Tools for building AI Workflows

Creating an AI Workflow

Creating an AI Workflow
Using **n8n**



Building AI workflows

AI Workflows

By **specifying the actions**, we **reduce** the chances of the LLM generating **irrelevant** or **incorrect information**



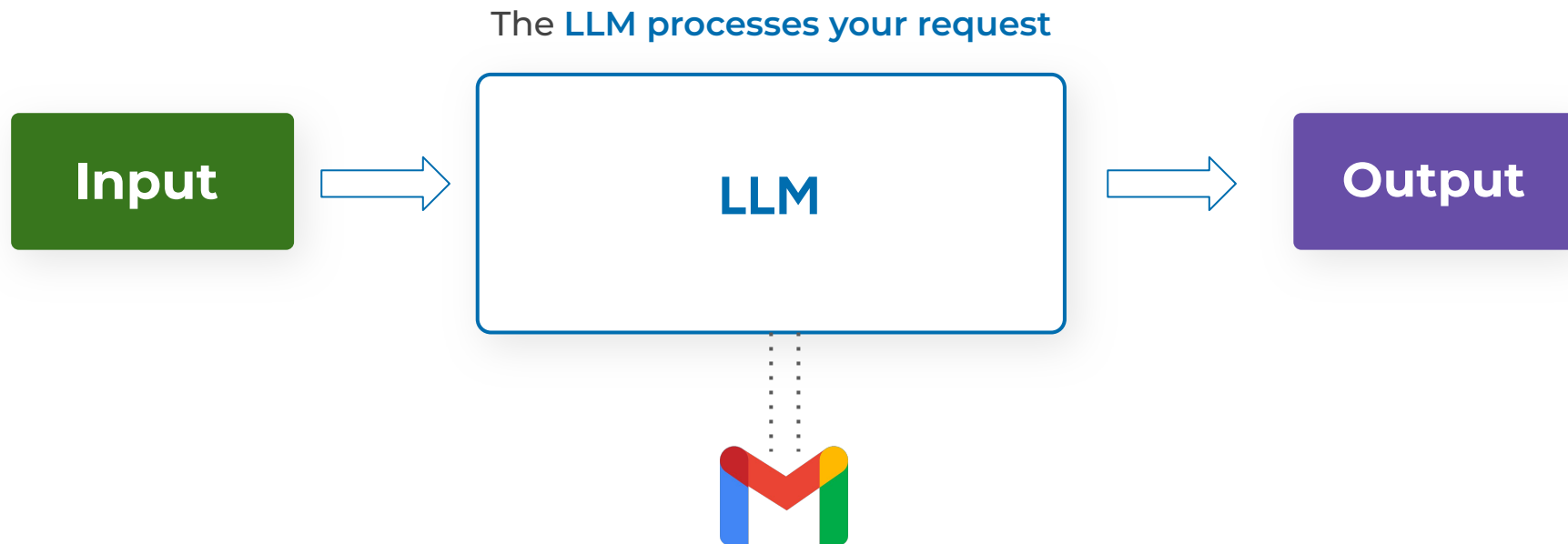


Creating AI Workflow using Make

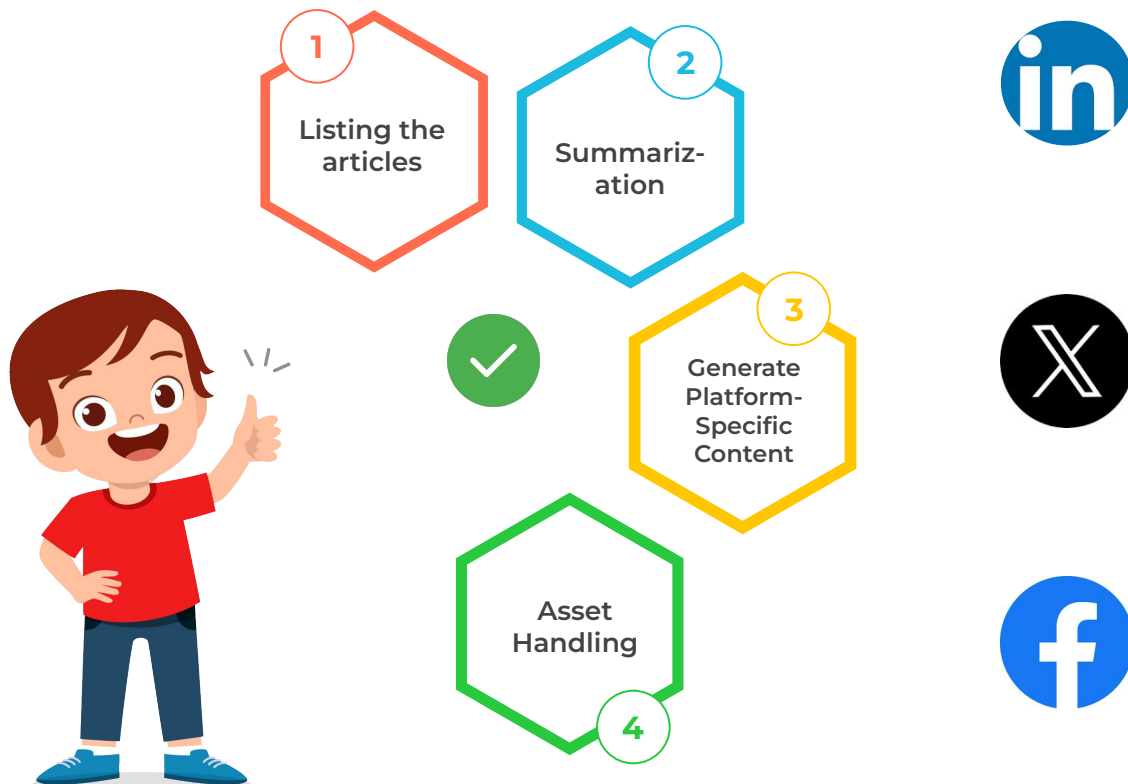
Hands on of Make.com

<https://www.youtube.com/watch?v=H0YRniHh2tg>

Example - Sending an Email



AI-Powered Social Media Content Creator & Publisher



Workflow

A workflow is a **pre-defined sequence** of steps for completing a task



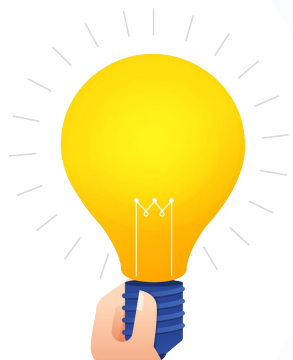
Example



Imagine you want to Build a social media manager that "**Summarizes an Article and Post it on LinkedIn, X(twitter), and Facebook**"



Quick Question



Can LLM do this?



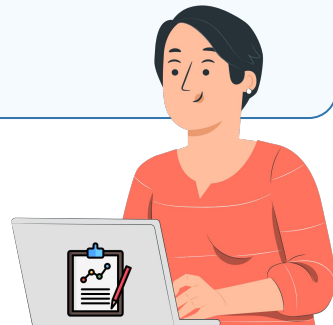
Setting the Scheduler

Explore and **try to run your workflow every 15 minutes.**

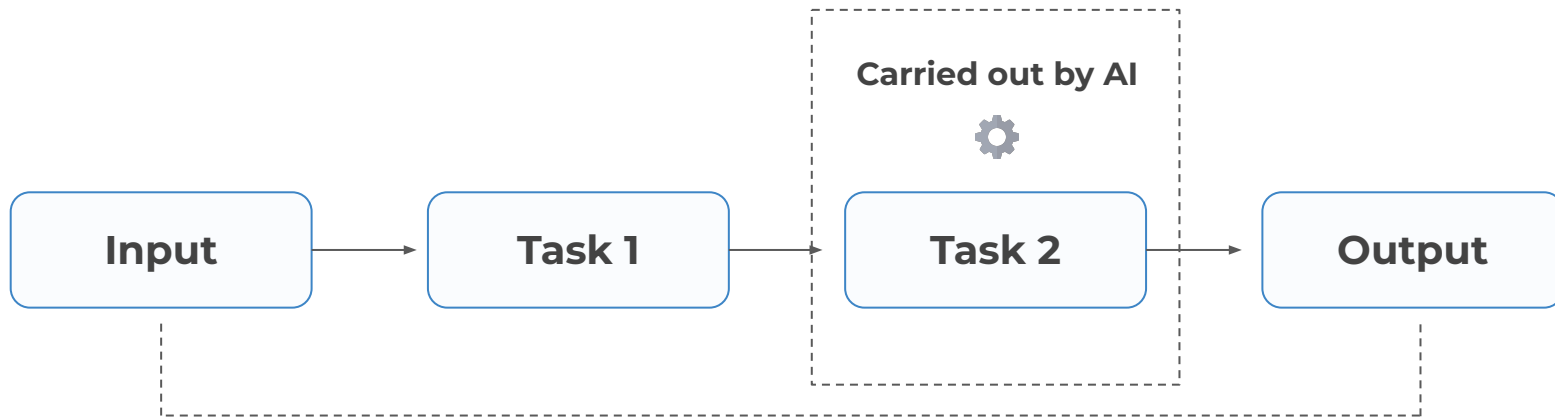


Prompt for Summarizing the Content

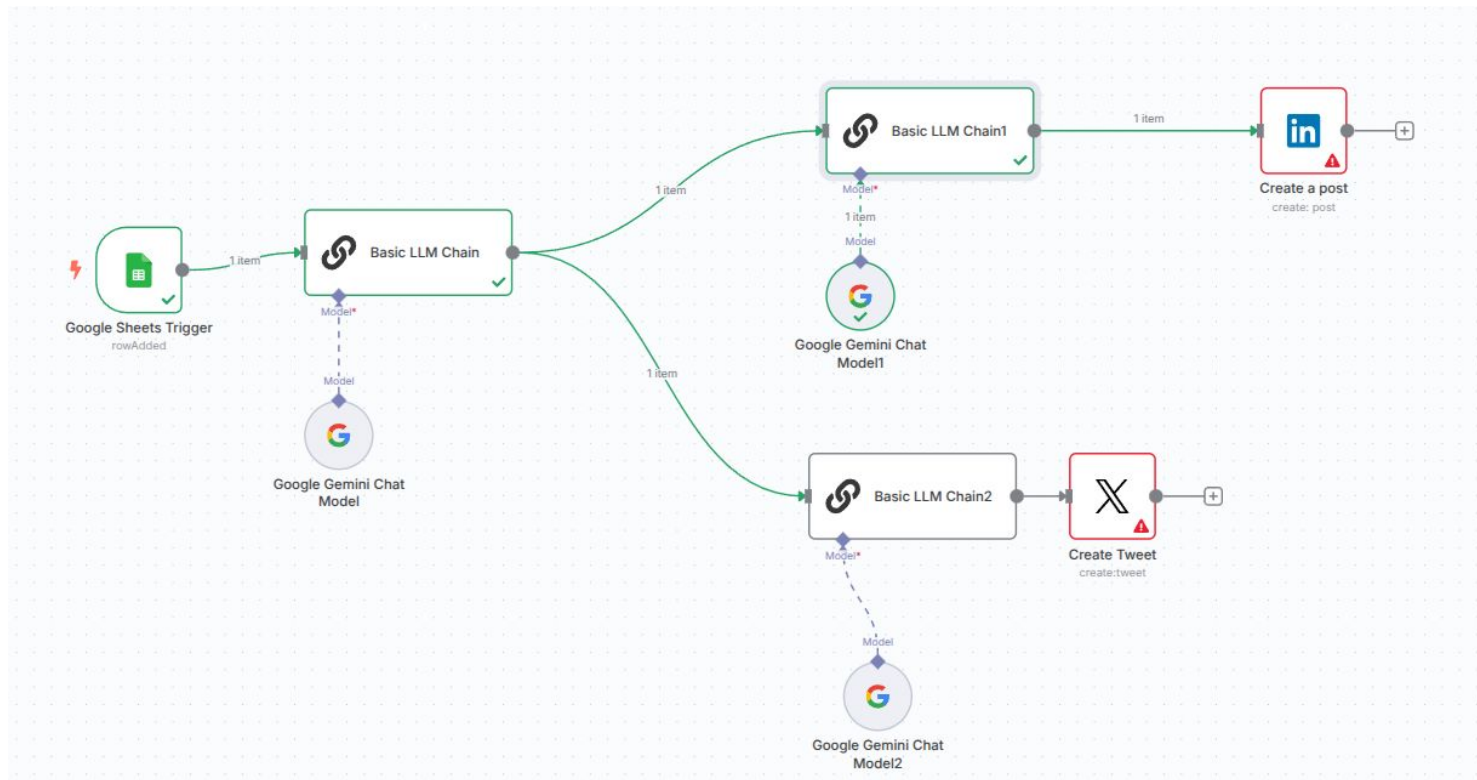
Assume the role of an Industry expert and draft a LinkedIn post discussing key points from a news article relevant to artificial intelligence. The post should provide insightful analysis, connect with current industry trends, and encourage professional engagement or discussions. Highlight any notable implications for the industry. LinkedIn supports detailed posts, so ensure it is informative and structured. Add a professional call to action. Here's a summary of the article:



Automated Workflow



For Instructor



Note



402 Errors

Content length exceeds the maximum characters allowed



Authentication Failures

API credentials expired or incorrect